Low-Level Design Handbook — Interview Edition

The book you re-read the night before an LLD / OOD or machine-coding round: all 22 chapters of the *Low-Level Design Handbook* (Java 21) cut down to the rules, tables, questions and traps that interviews test. Nothing here is new. Every line comes from the handbook; open the matching chapter there (Study or Full Edition) when you need the explanation, the diagram or the runnable code.

How to use this edition

- **One section per chapter (1–22)**, always in the same order: **Must-know** (one-line rules) → **Key facts & numbers** (tables) → **Code to remember** (short Java sketches) → **In the interview** (likely questions, short answers) → **Gotchas** (mistake → correct idea) → **TL;DR**.
- **Design patterns (Ch 8–11):** every GoF pattern and every "beyond GoF" extra is **one table row**: intent → when → structure hint → the handbook's Java example → pitfall. Ten patterns also get a ≤10-line Java sketch.
- **Case studies (Ch 14–21):** each of the 45 problems is a **compact card**: requirements → core classes → key patterns → tricky part. Learn the tricky part first; that is where interviewers spend their follow-up questions.
- **Night before (about 2 hours):** Top 25 questions below → Ch 1 (framework) → the pattern tables → Ch 12 recipe → three case cards you have not practised (say requirements, classes and the hard method aloud, then check the card) → Last-minute checklist. Then stop; sleep beats one more pattern.
- **Morning of (15 minutes):** framework (Ch 1) → SOLID table (Ch 4) → "Which pattern?" table (Ch 11) → concurrency recipe (Ch 12) → the case cards you expect.
- **Rule for everything below:** recall first, then check. Recall is what the interview tests.

Top 25 questions across the book

#	Question	One-line answer	Ch
1	How do you approach an LLD question?	Clarify → requirements (8–12 + out-of-scope) → use cases → entities → relationships → class diagram → key flows → patterns/SOLID → code → edge cases/concurrency → extensions	1
2	LLD vs system design (HLD)?	HLD = many services and machines at scale; LLD = classes, relationships and code inside one application	1
3	When is something a class, not a field?	Different **behaviour** → class; different **data** only → field or enum	1, 3
4	Four pillars of OOP?	Encapsulation (data + rules together), abstraction (hide how), inheritance (true is-a only), polymorphism (one call, per-object behaviour; kills type switches)	2
5	Abstract class vs interface?	Interface = "can do" contract, many per class, no state; abstract class = "is a" with shared state/constructors, one parent	2
6	Overloading vs overriding?	Compile time by declared parameter types vs run time by the actual object; fields and statics are not polymorphic	2
7	`equals` / `hashCode` contract?	Equal objects must have equal hash codes; override both or hash collections break	2
8	Aggregation vs composition?	Both "has-a"; aggregation parts outlive the whole (hollow ◇), composition parts die with it (filled ◆); diamond at the **whole**	3
9	Explain SOLID with one example each	SRP: one actor per class; OCP: `switch(vehicleType)` → `ParkingRate` classes; LSP: Square/Rectangle; ISP: `Robot` can't `eat()`; DIP: service owns `OrderSaver`, `main` wires MySQL	4
10	DIP vs DI vs IoC?	Principle (depend on abstractions) vs technique (pass dependencies in) vs broad idea (framework calls your code)	4
11	When would you *not* apply SOLID / a pattern?	Nothing has varied yet, one implementation, throwaway code; rule of three; YAGNI decides *when* to cut the seam	4, 5, 8
12	Composition over inheritance — why?	Avoids fragile base classes and refused methods; allows run-time swapping	2, 5
13	Law of Demeter?	Talk only to immediate friends; `a.getB().getC().do()` → an intent-revealing method on `a`	5
14	Name some code smells and fixes	Long Function → Extract Function; Repeated Switches → Replace Conditional with Polymorphism; Feature Envy → Move Function; Refused Bequest → Replace Subclass with Delegate	7
15	Which patterns come up most?	Strategy, Factory, Observer, State, Singleton, Builder — each only when its signal appears	8
16	Thread-safe Singleton?	Eager, holder class, or `enum` (survives reflection + serialization); DCL needs `volatile`; prefer a DI singleton scope	8
17	Strategy vs State?	Same diagram; strategies are chosen from outside and never know each other; states know each other and advance the machine	11
18	Decorator vs Proxy vs Chain of Responsibility?	Decorator adds behaviour, every layer runs; Proxy controls access; in CoR one handler can stop the flow	9, 10
19	Adapter vs Facade vs Bridge?	Adapter converts one interface after the fact; Facade simplifies many objects; Bridge splits two dimensions up front	9
20	Observer vs Mediator vs Pub-sub?	Observer broadcasts, listeners decide; Mediator coordinates peers centrally; pub-sub is Observer through a broker (durability, retries)	10, 11, 20
21	How do you prevent double booking?	Name the real unit of inventory (show seat, room-night, vehicle-day) and make "if free then take" one atomic step: lock, CAS, or conditional `UPDATE`; holds with TTL	12, 15

#	Question	One-line answer	Ch
22	Deadlock: conditions and fix?	Mutual exclusion, hold and wait, no preemption, circular wait; fix = global lock ordering (or `tryLock` + back-off)	12
23	Why is `volatile` not enough for `count++`?	Visibility, not atomicity; use `AtomicInteger`/`LongAdder` or a lock	12
24	LRU cache in O(1)?	`HashMap` key → node + doubly linked list with sentinels; move to front on get; evict `tail.prev` from list **and** map	20
25	How do you run a 90-minute machine-coding round?	Packages model/repository/service/strategy/exception/driver; mandatory features in order; run every ~15 min; services return data, driver prints and continues	21

1. The LLD interview & how to approach it

What the round is, how it is judged, and the eleven-step framework you say out loud.

Must-know

- **LLD round** — design the classes of *one* application and code the main flows. **Machine coding** — the same, but the code must *run* (≈90–120 min, in-memory).
- **Five core steps** — requirements → actors & use cases → classes → relationships → code. The full framework adds class diagram, key flows, patterns/SOLID, edge cases/concurrency, extensions.
- **Four requirement questions** — Who uses it? What can each actor do? What are the limits? **What is out of scope?** (the only question that makes the problem smaller).
- **Checkable requirement** — has a **number** ("at most five books") and an **actor** ("a member can reserve"): the number becomes a constant + one check, the actor names the owning class.
- **Finding classes** — nouns → candidate classes; verbs → methods; a verb with no home means a missing class.
- **Five cut tests** — a value? another name for the same thing? changing state? does anything call it? does **behaviour** or only **data** differ? → *data → field/enum; behaviour → class*.
- **One sentence per class, no "and"** — `ParkingLot` "issues tickets **and** tracks fullness" = two jobs; name the tension aloud.
- **Information Expert** — put each method where its data is ("find a free spot" → `ParkingFloor`).
- **Classes beginners forget** — the **record-keeping** class (ticket, loan, booking) and the **rule** class (rate, strategy).
- **Code skeleton** — Models → Repositories → Strategies → Services → Driver.

Key facts & numbers

Step	Ask yourself	Output
1 Clarify	Who uses it? One site or many?	Shared understanding
2 Requirements	Functional? Non-functional (thread-safe, extensible, in-memory)? Out of scope?	8–12 numbered lines + out-of-scope list
3 Use cases	What can each actor do?	Actor → use-case list
4 Entities	Which nouns hold state or behaviour?	Classes, one sentence each
5 Relationships	Uses / knows / has / owns / is-a? How many?	Typed lines with multiplicity
6 Class diagram	Fields, key methods, interfaces, enums	UML class diagram
7 Key flows	2–3 main use cases; lifecycle of the core object	Sequence / activity / state diagram
8 Patterns & SOLID	Rule that varies? Lifecycle? Many listeners?	1–2 named patterns + one rejected
9 Code	Interfaces, enums, models, services, driver	Running code for main flows
10 Edge cases & concurrency	Bad input? Full? Two users, one seat?	Validation, exceptions, locks
11 Extensions	"What if X is added?"	Short answers, maybe one class

45-min OOD round	Time	90-min machine coding	Time
Clarify requirements	5–8 min	Read, requirements, assumptions	5–10 min
Actors and use cases	3–5 min	Model: entities, enums, interfaces	10–15 min
Find the classes	8–10 min	Core flow end to end	30–40 min
Connect the classes	8–10 min	Remaining features, validation	15–20 min
Write the code	12–15 min	Polish, one extension, demo	5–10 min
		Buffer	5 min (never zero)

Rule of thumb: **about a quarter of the time goes before you name a class**; in machine coding, **something runs by the halfway mark**.

Code habit interviewers notice	Instead of
Pass time in (`Instant now`, injected `Clock`)	Reading the clock inside logic
Money as `BigDecimal` (from strings) or whole cents in a `long`	`double` (0.1 + 0.2 ≠ 0.3)
`IllegalArgumentException` / `IllegalStateException` / custom exception	Silent failure, error codes
`Optional` when "not found" is normal	Returning `null`
`private final` fields	Public mutable fields
Domain names (`ParkingTicket`)	`Manager`, `Helper`, `Data2`

What the interviewer judges: (1) did you bound the problem? (2) one job per class? (3) right relationships? (4) code matches diagram? (5) can you explain one trade-off (choice + rejected option)? Machine coding adds: runs, modular, readable, handles bad input, thread-safe shared state (senior).

Code to remember

```
// The parking-lot core in one breath: fit rule on the enum, ticket records, rate is a rule class.
enum VehicleType {
    MOTORBIKE(List.of(SpotType.MOTORBIKE, SpotType.COMPACT)),
    CAR(List.of(SpotType.COMPACT, SpotType.LARGE)), TRUCK(List.of(SpotType.LARGE));
    private final List<SpotType> fits;
    VehicleType(List<SpotType> fits) { this.fits = fits; }
    boolean fits(SpotType s) { return fits.contains(s); } }
record ParkingTicket(String id, String plate, ParkingSpot spot, Instant entry) {}
interface ParkingRate { BigDecimal fee(Duration stay); }        // rule class -> Strategy
```

In the interview

Q: Give the 30-second approach. **A:** "I clarify and write 8–12 requirements with an out-of-scope list, list actors and use cases, pick classes from the nouns with one job each, connect them in a class diagram, sketch the key flow, run a SOLID check and pick one or two patterns where the signal is clear, then code the core flows, handle edge cases and concurrency, and finish with extensions and trade-offs."

Q: Why not just start designing? **A:** Without a boundary the problem grows forever and every omission becomes indefensible.

Q: The interviewer says "your choice". **A:** Pick the simpler option, say it, and write it down as a visible assumption.

Q: You won't finish — what matters most? **A:** Bounded scope, clean responsibilities, correct relationships, code that matches the diagram, one defended trade-off.

Q: How many patterns in 45 minutes? **A:** One or two where the signal appears, and name one you rejected.

Q: Make "users can book some seats" checkable. **A:** "A customer can hold at most 10 seats per show; unpaid holds release after 10 minutes."

Gotchas

- Coding in minute two → spend 8–13 minutes on requirements and actors first.
- No out-of-scope list → write it and get agreement; declining an idea politely is a **strong** move.
- One subclass per value (`CompactSpot`, `LargeSpot`) → one `ParkingSpot` with a `SpotType` enum.
- God class (`ParkingLotManager`) → one sentence per class; split on "and".
- Inheritance for reuse ("Car extends Engine") → composition.
- Pattern stuffing → one or two, each with its signal.
- Twelve classes at equal depth → two or three flows done fully.
- Ignoring two users on one resource → lock or atomic operation on the shared state.
- Non-running machine-coding code → thin end-to-end path running by halfway.

TL;DR

Bound the problem first (8–12 requirements + out of scope), different data → field / different behaviour → class, one sentence per class, SOLID as quick checks, one or two patterns with a signal; inject time, exact money, guard shared state.

2. OOP fundamentals

The four pillars and the Java tools behind them: interfaces, abstract classes, enums, records, equality, generics.

Must-know

- **OOP** — data bundled with the only code allowed to change it.
- **Encapsulation** — fields private; only the owning class's methods change them and they enforce the rules (no setters that break invariants).
- **Abstraction** — expose only what callers need; hide how (interfaces, small public API).
- **Encapsulation vs abstraction** — hiding *data* behind rule-enforcing methods vs hiding *complexity* behind a minimal set of operations.

- **Inheritance** — child reuses and specialises parent, **only when "child is a parent" is truly correct**.
- **Polymorphism** — one call, per-object behaviour; replaces `switch`/`instanceof` chains. **Dynamic dispatch** = JVM picks the override from the actual class at run time.
- **Overloading** (compile time, declared parameter types) vs **overriding** (run time, actual object). Fields and `static` methods are **not** polymorphic (statics are hidden).
- **`this` / `static` / `final`** — current object / one per class / cannot be reassigned, overridden or extended.
- **Interfaces** — contracts; may have `default`, `static`, `private` methods; **functional interface** = one abstract method (lambda target).
- **Enums with behaviour** — fixed sets with fields and methods ("which spot types fit a truck" lives on `VehicleType`).
- **Records** — compact immutable data carriers, but only **shallowly** immutable: copy collections in the compact constructor.
- **Immutable class** — final class, private final fields, no setters, validate, defensive copies in and out (or a record).
- **Access** — default to `private`; package-private = same package only.
- **Generics** — type-safe reusable infrastructure; **PECS**: producer `extends`, consumer `super`.
- **Coupling / cohesion** — low coupling between classes, high cohesion inside a class.
- **OOA → OOD → OOP** — what the problem is → how, with objects → code. Connections that carry data become objects.

Key facts & numbers

	Abstract class	Interface
Instantiable	No	No
Instance fields	Yes	No (only `public static final` constants)
Constructors	Yes	No
Method bodies	Any	`default`, `static`, `private` only
Per class	**One** (`extends`)	**Many** (`implements`)
Expresses	"is a" (shared identity + state)	"can do" (capability / role)
Typical use	Shared state + partial implementation (Template Method)	Contracts, roles, strategies, test seams

Rule of thumb: start with an interface; add an abstract class underneath when implementations share real state or a fixed algorithm (JDK: `List` + `AbstractList` + `ArrayList`).

Pillar	Java tools	Smell when missing
Encapsulation	`private` fields, no setters, validation in constructors	Public fields; setters breaking invariants
Abstraction	Interfaces, abstract classes, small API	Callers reach into internals
Inheritance	`extends`, `super`, `@Override`	Subclass refuses or throws on parent methods
Polymorphism	Overriding + dynamic dispatch	`switch` / `instanceof` chains on type

Code to remember

```
public record Money(BigDecimal amount, String currency) {          // value object
    public Money { Objects.requireNonNull(currency);
        if (amount.signum() < 0) throw new IllegalArgumentException("negative"); }
    public Money plus(Money o) { return new Money(amount.add(o.amount), currency); }
}
public record Order(List<String> items) {                          // shallow -> copy it
    public Order { items = List.copyOf(items); }
}
```

In the interview

Q: Pillars of OOP in 30 seconds? **A:** "Encapsulation keeps data and its rules in the owning class; abstraction exposes only what callers need; inheritance reuses a parent when is-a truly holds; polymorphism lets one call

behave per object, removing type switches. I prefer interfaces and composition, enums for fixed sets, records for values, and consistent equals/hashCode."

Q: How do you make a class immutable? **A:** Final class, private final fields, no setters, defensive copies (or a record that copies its collections).

Q: What is dynamic dispatch? **A:** The JVM choosing the override from the object's actual class at run time.

Q: Where does "which spots fit a truck" live? **A:** On the `VehicleType` enum.

Q: Is a class with `private final Date created` immutable? **A:** Not unless it copies the mutable `Date` in and out — prefer `Instant`.

Q: Interface or abstract class first? **A:** Interface; add an abstract class only for real shared state or a fixed algorithm.

Gotchas

- Overriding `equals` without `hashCode` → hash collections lose entries; override both from the same fields.
- `final List` field → the reference is final, the list is not; copy with `List.copyOf`.
- Expecting a static method or field to be overridden → statics are hidden; only instance methods dispatch dynamically.
- Inheritance to share code ("is-a" false) → composition.
- Public mutable fields → private + behaviour methods.

TL;DR

OOP = data + the only code allowed to change it. Pillars: encapsulation, abstraction, inheritance (true is-a), polymorphism (kills type switches). Interfaces for contracts, abstract classes for shared state, enums for fixed sets, records for values, `equals` + `hashCode` together, private by default.

3. Class relationships & UML

The six relationships (what each claims about lifetime and substitution) and the few UML diagrams worth drawing.

Must-know

- **Six relationships, weakest → strongest coupling** — dependency, association, aggregation, composition, realization, generalization.
- **Dependency** — A *uses* B (parameter, local, return, static call); dashed arrow.
- **Association** — A *knows* B (field, created elsewhere); solid line, arrow if one-way.
- **Aggregation** — A *has* B, parts **outlive** the whole; hollow ◇ at the whole.
- **Composition** — A *owns* B, parts **die with** the whole, not shared; filled ◆ at the whole.
- **Realization** — A *implements* B; dashed line, hollow △ at the interface.
- **Generalization** — A *is a* B; solid line, hollow △ at the parent.
- **Multiplicity changes code** — `1` required field; `0..1` `Optional`/nullable; `1..*` non-empty collection checked in the constructor; exact numbers checked in constructors.
- **Visibility** — `+` public, `-` private, `#` protected, `~` package.
- **Qualified association** — lookup by key (`Library [isbn] → Book`, i.e. a `Map`). **Association class** — data about the link itself (`Enrollment{grade}`).
- **Use case** «include» = always (base → included); «extend» = sometimes (extension → base).
- **Sequence arrowheads** — filled = synchronous, open = asynchronous, dashed = return.
- **State transition label** — `event [guard] / action`.

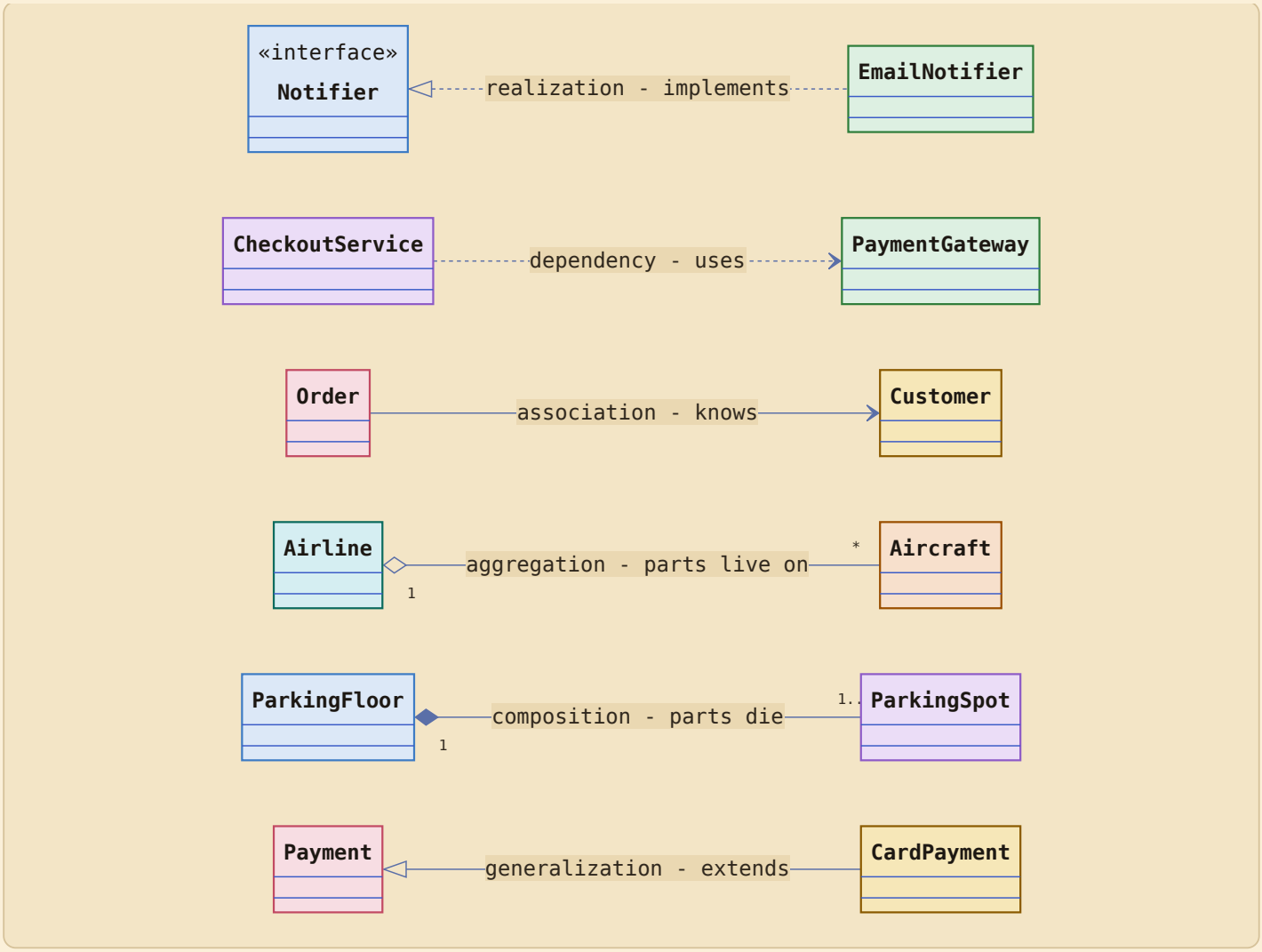
Key facts & numbers

Relationship	Phrase	UML line	Java shape	Lifetime	Example
Dependency	uses	dashed, open arrow	parameter / local	none	`CheckoutService` ⇢ `PaymentGateway`
Association	knows	solid line/arrow	field, created elsewhere	independent	`Order` → `Customer`
Aggregation	has, parts live on	solid, ◇ at whole	field/collection passed in	B outlives A	`Airline` ◇— `Aircraft`
Composition	owns	solid, ◆ at whole	A creates and hides B	B dies with A	`ParkingFloor` ◆— `ParkingSpot`
Realization	implements	dashed, △ at interface	`implements`	—	`EmailNotifier` ⇢ `Notifier`
Generalization	is a	solid, △ at parent	`extends`	—	`CardPayment` → `Payment`

Choosing, in order: interface it promises → realization; substitutable with no refused methods → generalization; not kept in a field → dependency; part dies with whole → composition; part outlives it → aggregation; otherwise → association.

Diagram	Use it for	Budget in 45 min
Use case	Scope: actors + what each does	Usually spoken, 3–5 min
Class	**The answer**: 10–15 boxes, every line typed, with multiplicity	Always drawn, 10–15 min
Sequence	Call order (3–4 objects, 6–8 messages)	One behaviour diagram only
Activity	A branching flow; fork/join for parallel steps	"
State machine	One object's lifecycle	"

UML has **14 diagram types**: 7 structural (class, object, package, component, composite structure, deployment, profile) and 7 behavioural (use case, activity, state machine, sequence, communication, interaction overview, timing).



Mermaid: `A ..> B` dependency · `A --> B` association · `A o-- B` aggregation, `A *-- B` composition (A is the whole) · `B <|.. A` realization · `B <|-- A` generalization.

Code to remember

```
class ParkingFloor {                                    // composition: floor creates and owns spots
    private final List<ParkingSpot> spots = new ArrayList<>();
    ParkingFloor(int n) { for (int i = 0; i < n; i++) spots.add(new ParkingSpot(i)); }
}
class Airline {                                         // aggregation: aircraft passed in, outlive airline
    private final List<Aircraft> fleet;
    Airline(List<Aircraft> fleet) { this.fleet = List.copyOf(fleet); }
}
```

In the interview

Q: Six relationships from weakest to strongest? **A:** Dependency, association, aggregation, composition, realization, generalization.

Q: Where does the diamond go and what does filled mean? **A:** At the whole; filled = composition (the part dies with the whole).

Q: `CompactSpot` / `LargeSpot` subclasses that only pass a constant to `super`? **A:** One `ParkingSpot` with a `SpotType` enum.

Q: If you draw only one diagram? **A:** The class diagram — it is the answer; test it with one behaviour diagram.

Q: Why is `Box → Piece` in chess not composition? **A:** A captured piece is not destroyed with the square.

Gotchas

- Diamond at the part end → it goes at the **whole**; triangle at the child → it goes at the **parent/interface**.

- Lines without type or multiplicity → label every line.
- Inheritance used to share data → one class + enum field.
- Drawing every diagram → class diagram always, one behaviour diagram at most.

TL;DR

Relationships claim lifetime and substitution. Choose by the substitution test, then "dies with the whole?", then "outlives?", else association; parameter-only = dependency. Multiplicity becomes code. Draw little, label everything, say the test aloud.

4. SOLID principles

Five names for the ways code becomes painful to change — a diagnostic tool, not a checklist.

Must-know

- **SRP** — one reason to change = **one actor** asks for its changes. Signal: responsibility sentence needs "and"; *Divergent Change*, *Large Class*.
- **OCP** — add behaviour by adding code, not editing working code. Signal: `switch`/`if` over a type. Closure is against *a kind of* change, not all editing.
- **LSP** — a subtype is usable anywhere its parent is: demands no more (preconditions not stronger), guarantees no less (postconditions not weaker), keeps invariants and history, throws nothing new.
- **ISP** — no client depends on methods it doesn't use → split into **role interfaces**, one per kind of caller.
- **DIP** — depend on abstractions; **the policy (high-level) owns the abstraction**; `main` wires the details.
- **SRP vs ISP** — SRP splits classes (guards the worker); ISP splits interfaces (guards the caller).
- **DIP vs DI vs IoC** — principle / technique (constructor injection) / broad idea (framework calls you; containers perform DI). A concrete parameter is injection without inversion.
- **Cohesion** — do the parts of this class belong together? **Coupling** — how much does A depend on B? **SoC** — is each area of work in its own module?
- **Order to apply** — S → O → L → I → D, each triggered by a real change.
- **YAGNI is the counterweight** — SOLID says *what* a good seam looks like; YAGNI says *when* to cut it.

Key facts & numbers

Principle	Smell that signals a violation	Fix	Patterns that help
S	"and" in the sentence; Divergent Change	Split by actor (Extract Class); names must stand alone	Command, Proxy, Decorator
O	Type `switch`; Repeated Switches, Shotgun Surgery	Varying part behind an interface; variants as classes	Strategy, Template Method, Decorator, Factory Method, Observer, Visitor
L	Subclass throws/no-ops a parent method; Refused Bequest	Better parent, narrower interface, composition	Composite, Null Object
I	Empty bodies / `UnsupportedOperationException`	Role interfaces per caller	Adapter
D	High-level class `new`s a low-level one; untestable	Inject an interface owned by the high-level class	Strategy, factories, Adapter, Observer

Over-engineering warning sign	Misapplied	Do instead
`XHelper`, `XUtils` after a split	SRP	Split by actor
Small classes always changed in the same commit	SRP over-split	Merge back
Interface + factory + registry for one implementation	OCP (Speculative Generality)	Write the `if`; refactor at case 2–3 (rule of three)
Subclass that silently does nothing	LSP	Better parent or composition
One interface per method	ISP over-split	Group by caller
Interface in front of `String`-like classes	DIP	Invert only volatile dependencies
DI container for a 200-line program	DIP	Wire by hand in `main`

Five-minute pass over your diagram: run S → O → L → I → D. **Two findings is a good pass; five means rework; zero means you didn't look.** S and O earn the most credit; O is what a late requirement probes ("trucks charged differently?" → "add a class").

Code to remember

```
// OCP + DIP: the policy owns the abstraction; variants are classes; main wires.
interface ParkingRate { BigDecimal fee(Duration stay); }
final class HourlyRate implements ParkingRate { /* tiers */ public BigDecimal fee(Duration d) { ... } }
final class ExitService {
    private final ParkingRate rate;                    // injected, not `new HourlyRate()`
    ExitService(ParkingRate rate) { this.rate = rate; }
    BigDecimal charge(ParkingTicket t, Instant now) { return rate.fee(Duration.between(t.entry(), now)); }
}
```

In the interview

Q: `switch (vehicleType)` computing fees? **A:** OCP — a `ParkingRate` interface, one class per rate.

Q: `ReadOnlyList extends ArrayList` throws on `add`? **A:** LSP; compose, or use a read-only interface.

Q: Why is Square/Rectangle worse than `Penguin.fly()` throwing? **A:** It fails **silently** (wrong area, no error).

Q: `Robot implements Worker { work(); eat(); }`? **A:** ISP; split by caller into `Workable` / `Eater`.

Q: `private final MySqlOrderDao dao = new MySqlOrderDao();`? **A:** DIP violation and untestable; inject an `OrderSaver` interface the service owns.

Q: How do you name a design problem aloud? **A:** Observation → cost → fix → price of the fix; describe before labelling ("`Reporting` implements nine methods but calls two — an interface segregation problem").

Q: Is SOLID still relevant / would you always apply it? **A:** As diagnostic vocabulary, yes; almost never "always" — flexibility later is bought with simplicity now.

Q: Most violated principle? **A:** SRP (responsibilities accumulate), or ISP in reverse (an interface per method).

Gotchas

- Reciting definitions → spot the problem in *this* design and state its cost.
- Applying all five at once → apply one when a real change asks for it.
- OCP "forbids editing" → no, it closes the code against one kind of change.
- "Composition always" → only when behaviour varies or a subclass would refuse a method.
- Confidently naming the wrong principle → describe first, label second.

TL;DR

SRP one actor per class; OCP varying part behind an interface; LSP keep every promise, compose when a subclass would refuse; ISP one role per caller; DIP policy owns the abstraction, `main` wires. Diagnose, don't checklist; always state what the fix costs.

5. More design principles (DRY, KISS, YAGNI, Demeter, GRASP…)

The principles that sit beside SOLID, and the tensions between them.

Must-know

- **DRY** — each piece of *knowledge* has one authoritative home (value objects, constants, methods).
- **KISS** — simplest design that works. **YAGNI** — don't build before it's needed (delete Speculative Generality).
- **Law of Demeter** — only talk to immediate friends; no train wrecks `a.getB().getC().do()`.
- **Tell, Don't Ask** — tell the owner what to do instead of pulling its data out and deciding for it (fixes anemic models, Feature Envy).
- **Composition over inheritance** — prefer has-a for reuse; avoids fragile base classes.
- **Program to an interface** — depend on supertypes; pick the concrete class once, at the edge.
- **Encapsulate what varies** — isolate the changing part behind an interface (Strategy, lambdas).
- **Loose coupling** — interacting objects know little about each other (Observer, events, DI).
- **Hollywood Principle / IoC** — "don't call us, we'll call you" (Template Method, callbacks, DI).
- **Least Astonishment** — do what the name promises, nothing more. **Fail fast** — guard clauses, `requireNonNull`.
- **Immutability** — objects never change after creation → no aliasing bugs, no races.
- **Design by Contract** — preconditions (caller's duty), postconditions (method's), invariants (always true).
- **GRASP** — nine guidelines for "which class should do this?" (table below).

Domain-Driven Design (§17) — must-know

- **Ubiquitous language** — the business's words in the code: `order.place()`, not `setStatus(2)`.
- **Bounded context** — one model + one vocabulary inside a boundary (Sales "order" ≠ Shipping "consignment"); often one team/service. Translate at the border with an **anti-corruption layer**.
- **Entity** = identity that survives change (equal by id); **value object** = defined by values, immutable, equal by value (`Money`, `Address`) — prefer value objects over primitives.
- **Aggregate** = cluster changed as one unit; outsiders touch only the **root**, which guards invariants. Keep it **small**, reference other aggregates **by id**, change **one aggregate per transaction**.
- **Domain events** (`OrderPlaced`) — past-tense facts; publish **after commit** (transactional outbox); other aggregates/contexts catch up eventually (sagas for multi-step flows).
- **Repository per aggregate root** (no `OrderLineRepository`); application services load → call one domain method → save.
- **Skip heavy DDD** for CRUD apps; value objects and good names are always worth it.

Key facts & numbers

GRASP guideline	Rule
Information Expert	The class with the data does the work
Creator	The container (or holder of the initial data) creates
Controller	A thin non-UI object receives each system event and delegates
Low Coupling	Pick the design with fewer, more stable dependencies
High Cohesion	Each class does one related set of things
Polymorphism	Each type implements the varying behaviour; no type switches
Pure Fabrication	Invent a helper (e.g. repository) when no domain class is a clean home
Indirection	A middleman between things that shouldn't know each other
Protected Variations	Wrap what's unstable behind a stable interface

Tension	Resolution
DRY vs loose coupling	Sharing code couples its users; share knowledge, not coincidence
YAGNI vs encapsulate-what-varies	Abstract only real or clearly predicted variation
Law of Demeter vs cohesion	Too many delegates → Middle Man smell
Information Expert vs SoC	Domain objects compute; fabrications (repositories) persist

Code to remember

```
// Tell, Don't Ask + Demeter: the owner enforces its own rule.
// Ask (bad): if (account.getBalance().compareTo(amt) >= 0) account.setBalance(account.getBalance().subtract(amt));
final class Account {
    private BigDecimal balance;
    void withdraw(BigDecimal amt) {
        if (amt.signum() <= 0) throw new IllegalArgumentException("amount");     // fail fast
        if (balance.compareTo(amt) < 0) throw new IllegalStateException("insufficient");
        balance = balance.subtract(amt);
    }
}
```

In the interview

Q: DRY — is duplicated code always bad? **A:** Duplicated *knowledge* is; two pieces that look alike but change for different reasons should stay apart (sharing them couples their users).

Q: Law of Demeter fix for `order.getCustomer().getAddress().getCity()`? **A:** Hide Delegate — an intent-revealing method such as `order.shippingCity()`; stop before it becomes a Middle Man.

Q: What is an anemic domain model? **A:** Classes with data but no behaviour, logic in services pulling the data out; fix with Tell, Don't Ask.

Q: DDD: entity or value object? **A:** Tracked over time by identity → entity (`Order` 1001); replaced when it changes → value object (`Money`, `Address`).

Q: How do two aggregates stay consistent? **A:** One aggregate per transaction; publish a domain event through an outbox; a handler or saga updates the other (eventual consistency).

Q: Which GRASP guideline justifies a repository? **A:** Pure Fabrication (plus Information Expert for the domain logic).

Q: Where does the Hollywood Principle appear? **A:** Template Method, framework callbacks, DI containers.

Gotchas

- DRY by merging coincidentally similar code → shared code couples; keep it apart.
- Abstracting "just in case" → YAGNI; wait for the second or third case.
- Delegate methods everywhere to satisfy Demeter → Middle Man.
- Contracts enforced by magic error values → checks at method entry/exit and exceptions.

TL;DR

DRY knowledge, KISS, YAGNI; talk to friends, tell don't ask; compose, program to interfaces, encapsulate what varies; fail fast; immutability; contracts; GRASP answers "who does this?" — Information Expert first.

6. Clean Code

Rules for readable, changeable code, plus review, error-handling and testing strategy.

Must-know

- **Names** — reveal intent; pronounceable, searchable; one word per concept; code reads like the requirement.
- **Functions** — small; do one thing; one level of abstraction; stepdown order; ≤ 2–3 arguments; **no flag arguments**; no hidden side effects; **Command–Query Separation**; exceptions, not error codes.

- **Comments** — say it in code; comment only *why*, warnings, legal text, public contracts; delete commented-out code.
- **Formatting** — newspaper order (headline first), related things close, one automated team style.
- **Objects vs data structures** — objects hide data and expose behaviour; data structures expose data; no hybrids; no train wrecks.
- **Error handling** — unchecked exceptions with context; wrap vendor exceptions; empty collections / `Optional` / special case instead of `null`; never pass `null`; always clean up.
- **Boundaries** — keep third-party code behind your own interfaces; learn it with tests.
- **Unit tests** — three laws of TDD; one concept per test; Given–When–Then; **F.I.R.S.T.** (Fast, Independent, Repeatable, Self-validating, Timely).
- **Classes** — small by responsibility, cohesive, open for extension, depend on abstractions.
- **Systems** — build objects in one place (**composition root**) and inject them; cross-cutting concerns in proxies/aspects.
- **Simple design (Beck)** — passes the tests → reveals intention → no duplication → fewest elements (*Clean Code* swaps the middle two).
- **Boy Scout Rule** — leave the code a little cleaner than you found it.

Key facts & numbers

Area	Rule of thumb
Concurrency	Separate concurrency code; share as little as possible; use the library; keep locks small; plan shutdown
Code review order	Correctness → design → readability → tests → security → performance → operability; automate style; small PRs
Error strategy	Classify; translate at boundaries; retry only transient + idempotent with backoff; one global handler that logs once
Testing strategy	Many unit, some integration, few E2E; pick the right test double; verify state over interactions; don't mock what you don't own
Test doubles	Dummy, stub, spy, mock, fake
Smells catalogue	C/E/F/G/J/N/T heuristics as review vocabulary

Code to remember

```
// Flag argument -> two intention-revealing methods; null -> Optional.
// Bad: void render(boolean isSuite)
void renderForSuite() { ... }
void renderForSingleTest() { ... }
Optional<User> findByEmail(String email) { return Optional.ofNullable(byEmail.get(email)); }
```

In the interview

Q: What makes a function clean? **A:** Small, one thing, one abstraction level, few arguments, no flags, no hidden side effects, either a command or a query.

Q: When is a comment good? **A:** When it explains *why*, warns, or documents a public contract — not when it narrates unclear code (that's a smell; extract a function).

Q: Return `null` or what? **A:** Empty collection, `Optional` for "not found", or a special-case object.

Q: What is F.I.R.S.T.? **A:** Fast, Independent, Repeatable, Self-validating, Timely.

Q: When should a service retry? **A:** Only transient failures on idempotent operations, with backoff; log once in one global handler.

Gotchas

- `process(data, true)` → flag arguments hide two functions.
- A method that returns a value *and* changes state → split (CQS).
- Leaking vendor exceptions → wrap at the boundary.
- Mocking third-party types → wrap them and mock your wrapper.

- Commented-out code "just in case" → delete it; version control remembers.

TL;DR

Intent-revealing names, small single-purpose functions, comments for *why*, exceptions over codes and nulls, third-party code behind your interfaces, F.I.R.S.T. tests, one composition root, Beck's four rules.

7. Refactoring

Changing structure without changing behaviour — the 24 smells and the moves that cure them.

Must-know

- **Refactoring** — change internal structure without changing observable behaviour, in small steps, tests as the safety net.
- **Two hats** — at any moment you are either adding function or refactoring, never both.
- **Characterisation test** — records what existing code does now, before you refactor it.
- **Code smell** — a surface symptom of a deeper design problem, not a bug by itself.
- **Smell families** — Bloaters, OO Abusers, Change Preventers, Dispensables, Couplers (+ 2nd-edition extras).
- **Refactoring toward patterns** — algorithm switch → Strategy; status checks → State; `null` checks → Null Object; same skeleton → Template Method; many constructors → Factory Method; ordered setup calls → Builder; undo/queue → Command; logging/caching mixed in → Decorator; polling → Observer; wrong interface → Adapter; feature combinations → Strategy/Decorator/Bridge; a pattern not earning its keep → remove it.

Key facts & numbers

Smell	Signal	First move
Mysterious Name	`data`, `proc()`, `Manager`	Change Function Declaration / Rename
Duplicated Code	Same structure in several places	Extract Function
Long Function	You want a comment to explain a block	Extract Function
Long Parameter List	Many same-typed params, boolean flags	Introduce Parameter Object, Preserve Whole Object
Global Data	Writable from anywhere (incl. mutable singletons)	Encapsulate Variable
Mutable Data	Shared data changed after creation	Encapsulate Variable
Divergent Change	One class changed for many reasons (SRP)	Split Phase, Extract Class
Shotgun Surgery	One change → edits in many classes	Move Function, Move Field
Feature Envy	Uses another class's data more than its own	Move Function
Data Clumps	Values always travel together	Extract Class, Introduce Parameter Object
Primitive Obsession	Money as `double`, status as `String`	Replace Primitive with Object
Repeated Switches	Same type switch in several places	Replace Conditional with Polymorphism
Loops	Loop that filters/maps/aggregates	Replace Loop with Pipeline
Lazy Element	Class/function not pulling its weight	Inline Function / Inline Class
Speculative Generality	"Might need someday" hooks	Collapse Hierarchy, Inline, Remove Dead Code
Temporary Field	Field set only sometimes	Extract Class, Introduce Special Case
Message Chains	`a.getB().getC().getD()`	Hide Delegate
Middle Man	Most methods just forward	Remove Middle Man
Insider Trading	Modules poke each other's internals	Move Function / Move Field
Large Class	God class	Extract Class
Alternative Classes, Different Interfaces	`sendMail` vs `dispatch`	Change Function Declaration
Data Class	Only fields + getters/setters	Encapsulate Record, Move Function
Refused Bequest	Subclass rejects inherited methods (LSP)	Replace Subclass/Superclass with Delegate
Comments	Comment as deodorant	Extract Function

Smell family	Principle violated
Divergent Change, Large Class	SRP
Repeated Switches, Shotgun Surgery	OCP
Refused Bequest	LSP
Global / Mutable Data	Encapsulation
Hard-wired `new` of concrete dependencies	DIP
Duplicated Code / Speculative Generality	DRY / YAGNI
Message Chains, Insider Trading	Law of Demeter
Feature Envy, Data Class	Tell, Don't Ask

Code to remember

```java
// Replace Conditional with Polymorphism (the most-asked move)
// Before: switch (type) { case CAR -> 20; case TRUCK -> 50; ... } repeated in 3 places
sealed interface Vehicle permits Car, Truck {
    int hourlyRate();
}
record Car(String plate) implements Vehicle { public int hourlyRate() { return 20; } }
record Truck(String plate) implements Vehicle { public int hourlyRate() { return 50; } }
```

In the interview

Q: What is refactoring, and what isn't? **A:** Behaviour-preserving structural change in small tested steps; adding a feature at the same time is not refactoring (two hats).

Q: Legacy code with no tests — first step? **A:** Characterisation tests that pin current behaviour, then small refactorings.

Q: A 200-line method? **A:** Extract Function on each commented block, Replace Temp with Query, Decompose Conditional; maybe Replace Function with Command.

Q: Which smell signals an OCP problem? **A:** Repeated Switches / Shotgun Surgery.

Q: Name smells that point to Law of Demeter violations. **A:** Message Chains and Insider Trading.

Gotchas

- Refactoring without tests → you cannot prove behaviour didn't change.
- Big-bang rewrites → small compiling steps, run tests after each.
- Refactoring *into* patterns by reflex → only when the smell is real; removing a pattern is also a refactoring.

TL;DR

Refactor = same behaviour, better structure, tiny tested steps, one hat at a time. Know the 24 smells by their signal and their first move; smells map to principles; patterns are refactoring targets, not starting points.

8. Design patterns: introduction & creational

What a pattern is, how to pick one, the anti-patterns, and the five creational patterns plus Simple Factory and Object Pool.

Must-know

- **Pattern** = name + problem + solution template + consequences. Value: shared vocabulary and proven trade-offs. Risk: overuse.
- **GoF count** — 5 creational + 7 structural + 11 behavioral = **23**. Class-scoped (inheritance): Factory Method, class Adapter, Template Method, Interpreter; the rest are object-scoped (composition).
- **Pick from a signal, not from memory** — "a rule that varies", "creation depends on a type value", "many listeners", "a lifecycle", "exactly one", "many optional fields".
- **The six in almost every round** — Strategy, Factory, Observer, State, Singleton, Builder.
- **Anti-patterns** — God Object; Golden Hammer (favourite pattern everywhere); patternitis (factory for one class); Singletonitis (singletons as globals); cargo-cult structure without intent.
- **Singleton** — one instance **per JVM (strictly per class loader)**; eager, holder class, or `enum` (most robust against reflection and serialization); DCL needs `volatile`; it is global state → prefer a DI singleton scope. Many case studies reject it.
- **Simple Factory** — one `create(type)` with a switch; an idiom, **not** GoF, not strictly OCP (a `Supplier` registry makes it open).
- **Factory Method vs Abstract Factory** — one product varied by **inheritance** (a protected hook the base procedure calls) vs a **family** of matching products by **composition** (one factory object, chosen once at the composition root).
- **Builder** — required values in the builder constructor, named optional steps returning `this`, one validating `build()` → immutable product via a private constructor. Director optional; staged builder enforces required steps at compile time.

- **Prototype** — copy a configured instance; decide **shallow vs deep per field**; copy constructors, not `Cloneable`; never copy identity (DB id, connection).
- **Object Pool** — borrow/return a bounded set of expensive objects; release in `finally`, reset, validate; use HikariCP in production.

Key facts & numbers

Pattern	Intent	When	Structure hint	Handbook Java example	Pitfall
Singleton	Exactly one instance + global access	Genuinely one (config, registry)	Private ctor + static accessor (holder / enum)	`EagerConfigStore`, `EnumConfigStore`, `AppLogger`; `DiAlternative`	DCL without `volatile`; hidden global state; `reset()` "for tests"; "one per system" (it's per JVM)
Simple Factory (idiom)	Map a type value to a class	Small stable set of types	Static `create(type)` returning an interface	`NotificationFactory.create(Channel)`	Called "Factory Method"; `default` returning `null`; grows into a God class
Factory Method	Subclasses decide which product to create	Shared procedure, varying product	Abstract creator, `protected abstract createX()` called by the template	`ReportGenerator.createDocument()`; `SupplierReportGenerator`	Public factory method; calling it from the base constructor
Abstract Factory	Create matching families	Products must match; swap a whole set	Interface with one `createX()` per product	`StorageFactory` → `Connection` + `QueryBuilder`	New product kind edits every factory; choosing the factory deep in the call stack
Builder	Build complex objects step by step	~6+ args, many optional, cross-field rules	Static nested `Builder`, fluent setters, validating `build()`	`HttpRequest.Builder`, `Email`; `TelescopingPizza` (the problem)	No validation in `build()`; sharing mutable collections; public product ctor
Prototype	New objects by copying	Setup expensive; only the interface known	`copy()` / copy constructor; registry of prototypes	`NotificationTemplate`, `ShapeRegistry`; `CloneablePitfall`	Shallow-copied mutable list; trusting `super.clone()`; handing out the registry original
Object Pool (extra)	Reuse bounded expensive objects	DB connections, limited resources	`Semaphore` permits + idle queue + factory/reset/validate	`ObjectPool<T>`	Not releasing in `finally`; no reset; double release raising the cap

Question you're asking	Pattern
"Which class, from a type value?"	Simple Factory / Factory Method
"Which *family* of classes?"	Abstract Factory
"Too many constructor arguments?"	Builder
"Another one of *this*, cheaply?"	Prototype
"Only one, ever?"	Singleton (or DI singleton scope)
"Expensive, created and destroyed repeatedly?"	Object Pool

Code to remember

```
public final class ConfigStore {                    // Singleton: holder idiom
    private ConfigStore() { }
    private static final class Holder { static final ConfigStore INSTANCE = new ConfigStore(); }
    public static ConfigStore get() { return Holder.INSTANCE; }   // lazy + thread-safe via class loading
}
// public enum ConfigStore { INSTANCE; ... }          // survives reflection + serialization
```

```
public final class HttpRequest { private final String url, method, body;       // Builder
    private HttpRequest(Builder b) { url = b.url; method = b.method; body = b.body; }   // private: only build()
    public static final class Builder {
        private final String url; private String method = "GET"; private String body;
        public Builder(String url) { this.url = url; }                      // required -> constructor
        public Builder method(String m) { method = m; return this; }
        public Builder body(String b) { body = b; return this; }
        public HttpRequest build() {   // one validating exit; cross-field rule
            if (method.equals("GET") && body != null) throw new IllegalStateException("GET cannot have a body");
            return new HttpRequest(this); } } }
```

```
public abstract class ReportGenerator {               // Factory Method
    protected abstract Document createDocument();     // the hook (protected, not public)
    public final String generate() {                  // procedure written once
        List<String> rows = List.of("north,120", "south,95");
        Document d = createDocument(); d.render(rows); return d.fileName();
    }
}
class PdfReportGenerator extends ReportGenerator { protected Document createDocument() { return new PdfDocument(); }
}
```

In the interview

Q: Why does double-checked locking need `volatile`? **A:** Otherwise the reference can be published before the constructor's writes are visible; `volatile` adds the happens-before edge.

Q: Unique IDs across 10 servers — Singleton? **A:** No, a Singleton is one per JVM. Use a DB sequence, Snowflake-style IDs (timestamp + machine id + counter) or UUIDs.

Q: Simple Factory vs Factory Method? **A:** One static switch that knows every type vs subclasses overriding a hook the base procedure calls. Raise this yourself — calling a switch a Factory Method is a classic slip.

Q: Add SQLite to an Abstract Factory? **A:** `SqliteConnection`, `SqliteQueryBuilder`, `SqliteFactory`, one line in the composition root. Adding a new product kind (e.g. `MigrationRunner`) edits every factory — the weak spot.

Q: How does a Builder enforce required fields? **A:** Builder constructor, validation in `build()`, or a staged builder. The immutable product is thread-safe; the builder is not.

Q: Deep copy with cycles? **A:** Identity map original → copy; reuse the copy if already made.

Q: Static factory method vs constructor? **A:** Has a name, can cache (`Integer.valueOf`), can return subtypes.

Q: Pool sizing? **A:** From measured concurrency and downstream limits; small pools (≈ core-count scale) often win.

Gotchas

- Naive lazy Singleton in multithreaded code → holder / enum / eager.
- `Serializable` Singleton without `readResolve()` → second instance (enum avoids it).
- Singleton by reflex in case studies → inject one shared instance.
- Factory for a single concrete class → YAGNI; add when the second variant arrives.
- Director when the order never varies → skip it.
- `Cloneable` → copy constructor.

TL;DR

Patterns come from signals; use one or two and name one you rejected. Singleton: eager/holder/enum, prefer DI. Factory Method = inheritance hook for one product; Abstract Factory = matching families; Builder = named steps + validating `build()` → immutable; Prototype = copy, shallow vs deep per field.

9. Structural patterns

How objects are composed: make it fit, add to it, guard it, simplify it, split it, nest it, share it.

Must-know

- **Memory hooks** — Adapter *make it fit* · Decorator *add to it* · Proxy *guard it* · Facade *simplify it* · Bridge *split it* · Composite *nest it* · Flyweight *share it*.
- **Adapter** — implements **your** interface, holds the vendor's object, translates inputs, outputs **and errors**; prefer object adapters (composition, works with final classes). Only at boundaries you don't control.
- **Bridge** — two varying dimensions → two hierarchies + one reference; multiplication becomes addition (3 × 3 = 9 classes → 3 + 3 = 6). Same code as Strategy, different intent (permanent dimension designed up front).
- **Composite** — leaf and container share one interface; the container's child list is typed as **the interface**; safe (add only on composite) vs transparent (add on the interface); guard cycles; cache big totals.
- **Decorator** — same interface, holds the interface, wraps to add behaviour; N features = N classes, stackable; **order matters** (retry closest to what fails, authorization outside any cache).
- **Facade** — one door into a subsystem; wraps **several** objects; subsystem stays public; sequences and owns cleanup but **never decides** business rules.
- **Flyweight** — split intrinsic (shared, immutable) from extrinsic (passed in) state; factory-owned thread-safe pool; measure first.
- **Proxy** — same interface, controlled access: virtual (lazy), protection, remote, caching, logging, smart reference. Static (hand-written) or dynamic (`java.lang.reflect.Proxy` + `InvocationHandler`); Spring AOP is proxies.

Key facts & numbers

Pattern	Intent	When	Structure hint	Handbook Java example	Pitfall
Adapter	Convert an interface into the one clients expect	Vendor/legacy API doesn't match	Implements target, wraps adaptee, translates errors	`StripeGateway`, `PayPalGateway` → `PaymentGateway`; `FahrenheitSensor`	Leaking vendor types/exceptions; `if (vendor == STRIPE)` in one adapter; mapping "pending" to success
Bridge	Abstraction and implementation vary independently	Two dimensions multiplying classes	Abstraction holds an implementor interface	`Notification` (Alert/Reminder/Digest) × `Channel` (Email/SMS/Push)	Built "just in case"; abstraction branching on implementor type
Composite	Treat leaves and trees uniformly	Part-whole trees	Container holds `List<Component>` and delegates	`FileSystemNode` / `FolderNode` / `FileNode`; `Menu` / `MenuItem`	Child list typed as the leaf; returning the mutable list; no cycle check
Decorator	Add responsibilities by wrapping	Optional features in combinations	Implements + holds the same interface, forwards	`UserRepository` + `LoggingRepository`, `CachingRepository`; `Beverage` + `Mocha`, `Whip`	Holding the concrete class (can't stack); cache not invalidated on writes; auth inside a cache
Facade	Simple entry to a complex subsystem	Callers repeat the same multi-step sequence	One class calling several services in order	`OrderFacade.placeOrder` (inventory, payment, shipping, email)	Business rules in the facade; forgetting the failure path (release reservation)
Flyweight	Share intrinsic state among many objects	Millions of near-identical objects	Immutable flyweight + factory pool; extrinsic state as parameters	`MarkerStyleFactory` → `MarkerStyle` shared by `Marker`s	Extrinsic state inside the flyweight; `HashMap` check-then-put pool; `==` on boxed values
Proxy	Stand-in that controls access	Expensive, remote, restricted, cacheable	Same interface; creates or guards the real object	`LazyReport`, `CachingRateProxy`, `RemoteRateProxy`, `LoggingHandler` (dynamic)	Non-`volatile` lazy field; expensive call looks cheap (N+1); Spring self-invocation

Look-alikes	Difference
Adapter vs Decorator	Adapter changes the interface, keeps behaviour; Decorator keeps the interface, adds behaviour
Decorator vs Proxy	Same structure; Decorator adds behaviour to an object it's handed (often stacked); Proxy controls access and often creates the real object
Adapter vs Bridge	Adapter is retrofitted after the fact; Bridge is designed in up front
Facade vs Mediator	Facade: one-way, subsystem unaware of it; Mediator: two-way among peers that know it
Flyweight vs Object Pool	Flyweights shared by many at once; pooled objects used by one borrower at a time

Code to remember

```
interface UserRepository { User find(int id); void rename(int id, String name); }
final class CachingRepository implements UserRepository {          // Decorator
    private final UserRepository inner;                             // holds the INTERFACE -> stackable
    private final Map<Integer, User> cache = new ConcurrentHashMap<>();
    CachingRepository(UserRepository inner) { this.inner = inner; }
    public User find(int id) { return cache.computeIfAbsent(id, inner::find); }
    public void rename(int id, String n) { inner.rename(id, n); cache.remove(id); }  // invalidate on write
}
// new LoggingRepository(new CachingRepository(new DatabaseUserRepository()))
```

```
final class StripeGateway implements PaymentGateway {             // Adapter: implements OUR interface
    private final StripeClient stripe;
    StripeGateway(StripeClient stripe) { this.stripe = stripe; }
    public Receipt charge(Money m) {
        try { return new Receipt(stripe.createCharge(m.cents(), m.currency())); }
        catch (StripeCardException e) { throw new PaymentFailedException(e.getMessage(), e); } // translate errors
too
    }
}
```

In the interview

Q: Why not change the other side instead of writing an Adapter? **A:** Say the constraint up front: vendor library, another team, many callers.

Q: Bridge vs Strategy? **A:** Same code; Bridge splits a permanent dimension up front, Strategy makes one algorithm replaceable.

Q: Composite — transparent or safe? **A:** Safe if the tree is built in one place; transparent if callers add to arbitrary nodes. New operations without touching nodes → Visitor.

Q: Why does `@Transactional` not work on a same-class call? **A:** `this.method()` never goes through the proxy. JDK proxies need interfaces; CGLIB subclasses (no `final`).

Q: Why is `Integer 128 == 128` false? **A:** `Integer.valueOf` only guarantees caching for −128..127; use `equals`.

Q: Thread-safe flyweight factory? **A:** `ConcurrentHashMap.computeIfAbsent` — creation runs at most once per key.

Q: Several facades? **A:** Yes — one per use case, or layered.

Gotchas

- Class adapter by default → object adapter.
- Decorator forgetting to forward a method → every interface method must delegate.
- Facade that becomes a god object → it sequences, never decides.
- Unbounded caching proxy → bound it; remote proxy in a tight loop → hidden network cost.
- ORM lazy proxies → N+1; fix with fetch joins, entity graphs or batch fetching.

TL;DR

Adapter translates at boundaries; Bridge turns multiplication into addition; Composite = one interface for leaf and tree; Decorator = wrap, same interface, order matters; Facade = one door, no rules; Flyweight = share intrinsic, pass extrinsic; Proxy = same interface, controlled access (virtual, protection, remote, caching).

10. Behavioral patterns (1): Chain of Responsibility, Command, Iterator, Mediator, Memento, Interpreter

Behavioral patterns turn hidden control flow into named objects: handlers, requests, cursors, coordinators, snapshots, grammar nodes.

Must-know

- **Chain of Responsibility** — handlers linked to a successor; each handles, rejects or passes on; the chain is data, **order matters**, always a **terminal handler**, never swallow a request.

- **CoR vs Decorator vs pipeline** — CoR handler may **end** the request; Decorator layers all run and add behaviour; a pipeline runs every step.
- **Command** — a request as an object: **invoker** triggers and records, **command** knows do/undo, **receiver** holds the state. Undo **restores captured state**; history is a stack; a new command **clears redo**; irreversible actions get **compensation**. `Runnable` + `ExecutorService` is the everyday form.
- **Iterator** — traversal in a separate object (`hasNext`/`next`), **position lives in the iterator**; implement `Iterable<T>` for for-each; know fail-fast vs snapshot, external vs internal.
- **Mediator** — peers talk through one hub holding interaction rules; links grow linearly (4 widgets: 6 links → 4); coupling **moves** into the mediator — split it early.
- **Memento** — the originator snapshots itself into a token only it can unpack; the caretaker stores without reading; restore is one originator-controlled call (all-or-nothing).
- **Command vs Memento for undo** — Command stores the **action** + a little state (cheap); Memento stores the **whole state** (always correct). Large documents: command per edit + periodic snapshots.
- **Interpreter** — one class per grammar rule, parse to an AST, evaluate recursively with a context; tiny DSLs only (big grammars → ANTLR + Visitor).

Key facts & numbers

Pattern	Intent	When	Structure hint	Handbook Java example	Pitfall
Chain of Responsibility	Pass a request along handlers until one handles/stops it	Several checks, any may reject	Abstract `Handler` with `next`; `handle()` returns or passes on	`AuthHandler` → `RateLimitHandler` → `CreateOrderHandler`; expense `Approver`; `FilterChain`; `LogHandler`	Swallowing bug (no `passOn`); no terminal handler; undocumented order
Command	Turn a request into an object	Undo/redo, queue, schedule, retry, log	`Command.execute()`, `UndoableCommand.undo()`, invoker history stack	`AddItemCommand`, `ApplyDiscountCommand` on `Cart`; `CommandHistory`; `MacroCommand`	Recomputing in `undo`; not clearing redo; capturing a mutable object; unbounded history
Iterator	Traverse without exposing storage	Walk a collection, tree, paged API	`Iterable<T>` returning a cursor object	`BinarySearchTree` (in-order), `PagedSearchResults` over `SearchApi`	Modifying mid-loop; `next()` returning `null`; lazy loop hiding network cost (N+1)
Mediator	Centralise peer communication	Tangled interactions; order of reactions matters	Components know only `Mediator`; hub decides	`ShippingFormMediator` (`CountrySelect`, `MethodSelect`, `PriceLabel`, `SubmitButton`); `ChatRoom`; `ControlTower`	Peers calling each other "just once"; computation inside the mediator; stringly-typed events
Memento	Capture/restore state without breaking encapsulation	Undo, checkpoints, rollback	Originator creates/consumes sealed snapshot; caretaker stores	`DeploymentConfig` + `DeploymentHistory`	Shallow copy of mutable internals; caretaker reading fields; unbounded history
Interpreter	Evaluate sentences of a small language	User-written rules/expressions	AST of `Expression` nodes + `Parser`	`Expression` + `Parser` (arithmetic with variables)	Re-parsing each time; wrong precedence/associativity (`8 - 3 - 2` = `(8 - 3) - 2`); no error positions

Code to remember

```
interface UndoableCommand extends Command { void undo(); }
final class ApplyDiscountCommand implements UndoableCommand {
    private final Cart cart; private final int percent; private int previousPercent;
    ApplyDiscountCommand(Cart cart, int percent) { this.cart = cart; this.percent = percent; }
    public void execute() { previousPercent = cart.discountPercent(); cart.setDiscount(percent); } // capture
    public void undo()    { cart.setDiscount(previousPercent); }                    // restore, don't recompute
}
// CommandHistory: execute -> push on undo stack + clear redo; undo -> pop, undo(), push on redo
```

In the interview

Q: Chain of Responsibility in 30 seconds? **A:** "Each check is a handler linked to its successor; the request moves down until one completes or refuses it. Only the wiring knows the order. Servlet filters and HTTP middleware are the everyday examples. I give every path an explicit outcome and end with a terminal handler."

Q: Command vs Strategy? **A:** Strategy = *how* a job is done, swapped by the caller; Command = *which* job, with arguments, kept to run later or reverse.

Q: Internal vs external iterator? **A:** External: the client steps and can stop or interleave. Internal: the collection loops and calls your function — simpler, parallelisable, harder to stop early.

Q: Iterator vs Visitor? **A:** Iterator = how you move; Visitor = what you do at each element.

Q: Mediator vs Facade? **A:** Facade is one-way in front of an unaware subsystem; Mediator is two-way among peers. Microservice orchestration is mediator-style; choreography is observer-style.

Q: Interpreter vs Visitor? **A:** Interpreter keeps evaluation **in** the nodes; Visitor moves operations **out**.

Gotchas

- Business logic in the invoker → invoker only triggers and records.
- Stateful `FilterChain` shared across requests → one chain position per request.
- Reusing a one-shot iterator/stream → create a new one.
- Snapshotting connections or locks in a memento → only plain state.
- A "tiny" DSL growing into a language → stop, use a parser generator.

TL;DR

CoR: someone may stop the flow. Command: request as object, undo restores captured state. Iterator: position in the cursor. Mediator: one hub, keep it small. Memento: sealed snapshot, caretaker can't read. Interpreter: grammar as classes for tiny DSLs.

11. Behavioral patterns (2) & beyond GoF

Observer, State, Strategy, Template Method, Visitor, the non-GoF glue every interview expects, and the "which pattern?" table.

Must-know

- **Observer** — subject keeps a list of listeners behind one interface and notifies them; **iterate a copy, catch per listener, always unsubscribe**; `java.util.Observable` is deprecated; use a broker for durability/retries/ordering.
- **State** — one class per state; the context forwards each action; each state decides legality **and its successor**. Enums or sealed interfaces for small machines; transition tables/libraries for big ones. Persist the **state name/enum**, not the object.
- **Strategy** — interchangeable algorithms behind one interface, **chosen from outside** (config, factory `Map<String, Strategy>`, DI). The context stops branching. Lambdas for one-liners, records when configured; keep them immutable.
- **Template Method** — fixed order in a `final` base method; abstract required steps, **hooks** with default bodies, private fixed steps; Hollywood principle. Inheritance-based: one step varies.
- **Visitor** — operations become visitor classes; elements just `accept(v)` → `v.visit(this)`: **double dispatch**. Cheap to add operations, expensive to add types (expression problem). Java 21: **sealed interface + exhaustive**

pattern-matching `switch` usually replaces it.

- **Three look-alike questions** — (1) Do alternatives reference each other? yes → State, no → Strategy (Bridge if chosen once at wiring). (2) Can one participant stop the flow? yes → CoR; all run → Decorator; all told, none stops → Observer. (3) Does a central object decide what happens next? yes → Mediator; each listener decides → Observer.
- **Beyond GoF** — Null Object, Dependency Injection, Repository, Specification, MVC, Event Sourcing (lite): the glue of real systems.

Key facts & numbers

Pattern	Intent	When	Structure hint	Handbook Java example	Pitfall
Observer	Notify many dependents of a change	Open-ended set of independent listeners	Subject holds `List<Listener>`; `notify` loops over a copy	`BuildPipeline` + `BuildListener` (`SlackNotifier`, `DeployTrigger`, `MetricsRecorder`); `WeatherStation`; `EventBus`	No unsubscribe (leak); one throwing listener kills the loop; notifying before state updates; relying on order
State	Behaviour changes with internal state	Many states; methods repeat a status `switch`	Context holds `State`; states return/set successor	`Subscription` (`Trialing`, `Active`, `PastDue`, `Cancelled`); `VendingMachine`; `TrafficLight`	Business logic in states; illegal actions silently ignored; storing state objects in the DB
Strategy	Interchangeable algorithms	Several ways to do one job, picked at run time	Context holds a strategy interface	`ApiClient` + `RetryPolicy` (`FixedDelay`, `ExponentialBackoff`)	Every call site still chooses; fat interface; strategies knowing each other (that's State)
Template Method	Fixed skeleton, subclasses fill steps	Long fixed flow, a few varying steps	`final` template + abstract steps + hooks	`ImportJob` (`CsvImportJob`, `JsonImportJob`); `CaffeineBeverage` (`Coffee`, `Tea`)	Non-final template; overridable step called from base constructor; public steps
Visitor	Add operations to a stable type set	Many new operations over fixed types (ASTs)	`accept(v)` on elements; one `visit` per type	`Resource` (`Server`, `Database`, `LoadBalancer`, `ResourceGroup`) + `MonthlyCost`, `SecurityAudit`; `ModernPlan` (sealed switch)	Visitor for two types; `default` in a sealed switch (kills exhaustiveness); composite `accept` forgetting children
Null Object	Do-nothing object instead of `null`	`null` checks for an optional collaborator	Implements the interface with neutral behaviour	`GuestCustomer` from `CustomerDirectory`	Hiding real errors; use `Optional` for lookups
Dependency Injection	Receive collaborators via constructor	Classes `new` their own dependencies	Constructor params typed as interfaces; wiring in `main`/container	`OrderService(MessageSender)`; `RecordingSender` for tests; tiny `Container`	Service Locator (class asks a registry — anti-pattern)
Repository	Collection-like interface over storage	Business logic mixed with storage	Interface + in-memory/DB implementations	`UserRepository` / `InMemoryUserRepository` used by `SignupService`	Leaking storage types; confusing with table-oriented DAO
Specification	Named, combinable business rule	Same rule in queries and validation	`isSatisfiedBy` + `and`/`or`/`not`	`ProductSpecs` over `Product`	Finder-method explosion instead
MVC	Separate model, view, controller	UI, input and rules tangled	Observer (model→view) + Strategy (controller) + Composite (views)	`TaskModel`, `TaskView`, `TaskController` (`EditingController`, `ReadOnlyController`)	Logic in the view

Pattern	Intent	When	Structure hint	Handbook Java example	Pitfall
Event Sourcing (lite)	Store events; replay for state	Full history, audit, time travel	Append-only `EventStore`; state = fold of events	`ShoppingCart` + `CartEvent` + `EventStore`	Treating it as a plain audit log

If you see / need…	Use	Not to be confused with
One operation, several rules chosen at run time (pricing, matching, routing)	**Strategy**	State, Template Method
Class to create depends on a type value	**Factory**	Abstract Factory
Products must match as a family	**Abstract Factory**	Factory Method
~6+ constructor args, many optional	**Builder**	—
Existing class has the wrong interface	**Adapter**	Facade, Bridge
Two dimensions multiplying classes	**Bridge**	Adapter
Part-whole tree treated the same	**Composite**	Decorator
Add behaviour in combinations	**Decorator**	Proxy
Every caller repeats the same five steps into a subsystem	**Facade**	Mediator
Lazy load, access control, remote, caching	**Proxy**	Decorator
Several checks, any may reject	**Chain of Responsibility**	Decorator
Undo/redo, queue, schedule, retry actions	**Command**	Strategy
Order of reactions among many objects matters	**Mediator**	Observer
Roll back without exposing internals	**Memento**	Command undo
One change → open-ended set of listeners	**Observer** (or broker)	Mediator
3+ methods repeat a status `switch`	**State** (enum + guarded method for 3–4 states)	Strategy
Shared long fixed flow, few differing steps	**Template Method**	Strategy
Many new operations over stable types	**Visitor** / sealed types + pattern matching	Iterator
`null` checks everywhere	**Null Object** / `Optional`	—
Same rule in queries and validation	**Specification**	—

Code to remember

```java
@FunctionalInterface interface RetryPolicy {                    // Strategy
    int STOP = -1;
    int delayBeforeAttempt(int attempt);
}
record ExponentialBackoff(int baseMillis, int maxAttempts) implements RetryPolicy {
    public int delayBeforeAttempt(int a) { return a > maxAttempts ? STOP : baseMillis << (a - 2); }
}
// ApiClient holds a RetryPolicy and never asks which one it is; a lambda also works: a -> STOP
```

```
interface VendingState {                                        // State
    default void insertCoin(VendingMachine m, int c) { System.out.println("rejected " + c); } // refuse by default
    default void select(VendingMachine m, String code) { System.out.println("ignored"); }
}
final class Idle implements VendingState {
    public void insertCoin(VendingMachine m, int cents) { m.addBalance(cents); m.setState(new HasMoney()); } //
picks successor
}
```

```
final class BuildPipeline {                                     // Observer
    private final List<BuildListener> listeners = new CopyOnWriteArrayList<>();
    void subscribe(BuildListener l) { listeners.add(l); }
    void unsubscribe(BuildListener l) { listeners.remove(l); }
    void finish(int build, String result) {                     // update state first, then notify
        for (BuildListener l : listeners) {
            try { l.onBuildFinished(build, result); } catch (RuntimeException e) { log(e); } // one bad listener
can't stop others
        } }
}
```

```
abstract class ImportJob {                                      // Template Method
    public final void run(String source) {                      // final: order can't change
        List<String> records = new ArrayList<>();
        for (String row : readRows(source)) records.add(parseRow(row));   // required steps
        if (shouldValidate()) validate(records);                // hook
        save(records); afterSave(records.size());               // fixed step, then hook
    }
    protected abstract List<String> readRows(String source); protected abstract String parseRow(String row);
    protected boolean shouldValidate() { return true; }         // hook with a default
}
```

In the interview

Q: Observer vs pub/sub? **A:** In Observer the subject holds observers directly; pub/sub puts a broker between them, adding durability and retries.

Q: State vs Strategy? **A:** Same diagram; states name each other and advance the machine; strategies are picked from outside and never reference each other.

Q: Racing "cancel" and "deliver" on an order state machine? **A:** Lock, or optimistic `UPDATE ... WHERE status = 'PLACED'` and check the row count.

Q: Strategy vs Template Method? **A:** Template Method swaps **one step** via **inheritance** (fixed at creation); Strategy swaps **the whole algorithm** via **composition** (run time, several axes). Prefer Strategy when both fit.

Q: Why can't you just call `visitor.visit(resource)`? **A:** With static type `Resource` the compiler can't choose `visit(Server)`; `accept` supplies the second dispatch.

Q: Which patterns have you used? **A:** Concrete ones with a trade-off: Strategy for payment methods via an injected map; Observer via Spring events, moved to Kafka for retries; State for orders with `WHERE status = ?` guarding races.

Q: When would you avoid patterns? **A:** Two branches and no third coming, or when a lambda or enum does the job.

Gotchas

- Strategy chosen by a `switch` at every call site → the branch moved, it didn't disappear; choose once (factory/DI).
- Stateful strategies shared across threads → keep them immutable.
- Observers calling back into the subject → cycles; changing the list during notify → `ConcurrentModificationException` (iterate a copy).
- 12+ states with successors hard-coded everywhere → transition table.
- Template method without `final` → subclasses can reorder or skip steps.

TL;DR

Observer broadcasts; Mediator coordinates; CoR lets someone stop. States know each other, strategies don't. Template Method = inheritance, one step; Strategy = composition, whole algorithm. Visitor = cheap operations, expensive types (Java 21: sealed + switch). Null Object, DI, Repository, Specification, MVC, Event Sourcing are the real-system glue.

12. Concurrency & multithreading for LLD

The toolbox for shared mutable state, and the recipe for making booking-style operations safe.

Must-know

- **Race condition** — the result depends on thread timing on **shared mutable state**. **Critical section** — code touching that state; must run one thread at a time.
- **Check-then-act** — read a condition, then act, as two steps: a race unless made atomic ("find free → assign").
- **Concurrency vs parallelism** — many tasks in progress (structure) vs executing at the same instant (hardware). 50,000 idle chat connections = concurrency.
- **Thread states** — NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, TERMINATED. `run()` runs synchronously on the caller; `start()` starts a thread.
- **Java Memory Model** — **happens-before** edges (unlock → lock, volatile write → read, `start`, `join`, program order, transitive) make writes visible.
- `volatile` — visibility, **not atomicity**: `count++` still races.
- `synchronized` — mutex + visibility; different objects = different monitors (don't block each other). **Reentrant**: `lock()` ×3 needs `unlock()` ×3.
- **CAS** — "set only if still equal to expected"; failed CAS = someone changed it, re-read and retry; **ABA** → `AtomicStampedReference`.
- **Condition waits** — always in a `while` (spurious wakeups, competing threads); `wait()` releases the monitor while sleeping; a signal before the wait is a **lost wakeup**.
- **Deadlock** needs all four **Coffman conditions**: mutual exclusion, hold and wait, no preemption, circular wait. Break any one; usual fix = **global lock ordering**.
- **Deadlock vs livelock vs starvation** — all wait (BLOCKED/WAITING) / all run (RUNNABLE) but none progress / some progress while one never does.
- **Thread-starvation deadlock** — tasks in a bounded pool wait for tasks queued behind them in the same pool.
- **Immutability & confinement** — the cheapest thread safety: nothing to lock.

Key facts & numbers

Primitive	Use for	Gotcha
`synchronized`	Atomic multi-step update	Don't lock on a changing reference, boxed value or string literal; no slow I/O inside; readers need the same lock
`ReentrantLock`	+ try/timed/fair locking, `Condition`s	`lock()` just before `try`, `unlock()` in `finally`; fairness is slower
`tryLock(timeout)`	Give up if busy; deadlock avoidance	Never `unlock()` after a failed `tryLock`; randomised back-off (livelock)
`ReentrantReadWriteLock`	Many readers, few writers	No read → write upgrade (self-deadlock); writer starvation → fair mode
`StampedLock`	Optimistic reads	**Not reentrant**; `validate` the stamp
`AtomicInteger` / `AtomicLong`	Lock-free counters, single-variable claims	`get()` then `set()` still races — `compareAndSet` / `updateAndGet`; no side effects in the lambda
`LongAdder`	Very hot counters	Use instead of `AtomicLong` under high contention
`Semaphore(N)`	Limit access to N; signalling	Release in `finally`, only if acquired
`CountDownLatch`	Wait for N events once	One-shot; `countDown()` in `finally`
`CyclicBarrier` / `Phaser`	N threads meet every phase	Fewer threads than parties → wait forever
`ConcurrentHashMap`	Shared map	`containsKey` + `put` races → `putIfAbsent` / `computeIfAbsent` / `merge`; no nulls; no slow work inside `compute`
`CopyOnWriteArrayList`	Read-mostly lists (listeners)	Every write copies
Bounded `BlockingQueue`	Producer–consumer	Unbounded = no back-pressure; one poison pill per consumer
`ExecutorService`	Bounded task pool	Call `shutdown()`; `newFixedThreadPool` queue is **unbounded**; `submit` exceptions lost unless `get()`
Virtual threads	Many blocking I/O tasks	Pinning inside `synchronized` (Java 21–23); heavy `ThreadLocal`s
`CompletableFuture`	Combine async results	Handle errors with `exceptionally`
`ThreadLocal`	Per-thread state	`remove()` in pooled threads

Coffman condition	How to break it
Mutual exclusion	Lock-free/immutable data, read locks
Hold and wait	Acquire everything at once, or `tryLock` and release all on failure
No preemption	Timeouts, leases that expire
Circular wait	**Global lock ordering** (lower id first, tie-breaker for equals)

The LLD concurrency recipe: (1) name the shared mutable state (free spots, seat → owner, stock); (2) state the invariant ("a seat has at most one booking", "stock ≥ 0"); (3) find the critical check-then-act; (4) pick the smallest atomic tool — one item: `queue.poll()`, `putIfAbsent`, `compareAndSet`; several all-or-nothing: per-item locks **in sorted order**; across servers: DB constraints/transactions, optimistic versioning, distributed lock; (5) choose granularity — per floor / show / SKU, not one global lock; (6) handle time — holds with expiry, timeouts, idempotent retries.

`ThreadPoolExecutor`: threads beyond `corePoolSize` are created only when the queue is full (and below max).

Code to remember

```
// Seat booking: all-or-nothing, locks in sorted order (no circular wait)
boolean book(List<Seat> seats, String user) {
    List<Seat> ordered = seats.stream().sorted(Comparator.comparing(Seat::id)).toList();
    ordered.forEach(s -> s.lock().lock());
    try {
        if (ordered.stream().anyMatch(s -> s.owner() != null)) return false;   // check...
        ordered.forEach(s -> s.setOwner(user));                                // ...and act, atomically
        return true;
    } finally { ordered.forEach(s -> s.lock().unlock()); }
}
```

```
// Lock-free spot claim: one winner per spot
if (spot.occupied().compareAndSet(false, true)) { issueTicket(spot); } else { tryNextSpot(); }
counts.merge(word, 1, Integer::sum);                 // atomic word count
```

In the interview

Q: Why does `count++` lose updates? **A:** Read-modify-write: two threads read the same value before either writes. Use `AtomicInteger` or a lock.

Q: Non-volatile `ready = true` after preparing data; another thread spins on it. Bugs? **A:** It may never see the flag, or see it with stale data.

Q: Why `while` around `wait()`? **A:** Spurious wakeups and competing threads; recheck the condition.

Q: Why is bare `notify()` risky? **A:** It can wake the wrong waiter, or be lost if sent before the wait; pair it with a state variable checked in a loop; prefer `notifyAll` or separate `Condition`s.

Q: Which Coffman condition does sorting seat ids break? **A:** Circular wait.

Q: Lock-free spot allocation across gates? **A:** Free spots in a concurrent queue/sorted set; `poll()` atomically (+ CAS on the spot).

Q: How is a deadlock detected? **A:** Thread dump (`jstack` / `jcmd`) prints "Found one Java-level deadlock"; `ThreadMXBean.findDeadlockedThreads()`.

Q: Why a bounded queue? **A:** A slow consumer pushes back instead of the queue growing until memory runs out.

Q: Readers must see a fully built snapshot — how? **A:** Publish an immutable snapshot through a `volatile` field (the volatile write happens-before later reads).

Q: Does your in-memory lock work with three app servers? **A:** No — use DB conditional updates/unique constraints, optimistic versions, or a distributed lock.

Gotchas

- One global `synchronized` on `ParkingLot.park()` → correct but serialises every gate; lock per floor/type or use CAS.
- Checking availability in one call and booking in another → one atomic step.
- Locking seats in click order → deadlock between two multi-seat bookings; sort first.
- Calling a listener or another object's synchronized method while holding your lock → notify after unlocking.
- Swallowing `InterruptedException` → rethrow or `Thread.currentThread().interrupt()`.
- Retrying `tryLock` in a tight loop → livelock; add jitter.

TL;DR

Races come from shared mutable state. Name it, state the invariant, make check-then-act atomic with the smallest tool, lock fine-grained and in a fixed order, notify outside locks, and say what changes with many servers.

13. Classic concurrency problems

Eleven small programs that exercise the primitives; know the primitive and the invariant for each.

Must-know

- **FooBar / Zero-Even-Odd** — semaphores as batons: exactly one permit circulates.
- **FizzBuzz multithreaded** — one monitor + `notifyAll`: threads wait for *different* conditions on one monitor, so only waking all guarantees the right one wakes (`notify()` → hang).
- **H2O** — two semaphores (2 H slots, 1 O slot) + a barrier; release slots after printing so the next molecule can't form early.
- **Dining philosophers** — ordered forks (the last philosopher takes fork 0 first) or an n−1 semaphore → no circular wait.
- **TTL cache** — `ConcurrentHashMap` + `compute` + conditional `remove(key, e)` so a value written between read and remove isn't deleted.
- **Striped map** — N locks, bucket → stripe by hash; the stripe must depend only on the hash so a key's lock doesn't change on resize (stripe count divides table length).
- **Bounded blocking queue** — lock + two `Condition`s (`notFull`, `notEmpty`); `signal()` is safe with separate queues, `notify()` on one monitor is not.
- **Bloom filter** — bits only go 0 → 1 (atomic OR); false positives possible, **no false negatives**; ≈ 9.6 bits per element for 1% false-positive rate.
- **Parallel merge sort** — Fork/Join with a threshold; subtasks touch disjoint ranges of a shared buffer.
- **Token bucket** — refill + take is one atomic step (lock, or CAS on one immutable `State(tokens, lastRefill)`); across servers: Redis + Lua.

Key facts & numbers

Problem	Key primitive	Invariant to state
FooBar	2 semaphores	Exactly one permit across both
Zero Even Odd	3 semaphores	Zero dispatches; one permit circulates
FizzBuzz	monitor + `notifyAll`	Predicates partition 1..n
H2O	2 semaphores + barrier	≤ 2 H and ≤ 1 O per round
Dining Philosophers	ordered locks / n−1 semaphore	No circular wait
TTL cache	`ConcurrentHashMap` + `compute` + conditional remove	Never return or keep expired data
Striped map	lock array	Each bucket belongs to exactly one stripe
Blocking queue	lock + 2 conditions	0 ≤ count ≤ capacity; wait in loops
Bloom filter	atomic OR	Bits only go 0 → 1
Merge sort	Fork/Join	Subtasks touch disjoint ranges
Token bucket	lock or CAS on immutable state	Refill + take atomic

Code to remember

```
final class BoundedQueue<T> { BoundedQueue(int cap) { this.cap = cap; }
    private final Deque<T> q = new ArrayDeque<>(); private final int cap;
    private final ReentrantLock lock = new ReentrantLock();
    private final Condition notFull = lock.newCondition(), notEmpty = lock.newCondition();
    void put(T x) throws InterruptedException { lock.lock();
        try { while (q.size() == cap) notFull.await(); q.add(x); notEmpty.signal(); } finally { lock.unlock(); } }
    T take() throws InterruptedException { lock.lock();
        try { while (q.isEmpty()) notEmpty.await(); T x = q.poll(); notFull.signal(); return x; }
        finally { lock.unlock(); } }
}
```

In the interview

Q: When is `signal()` safe instead of `signalAll()`? **A:** When each condition queue holds only threads waiting for the same thing and one change enables exactly one waiter.

Q: Why one immutable `State` in the lock-free token bucket? **A:** `tokens` and `lastRefill` must change together; one CAS on one reference updates both.

Q: Token bucket global across 10 servers? **A:** Store it in Redis and update atomically with a Lua script.

Q: Why `remove(key, e)` in the TTL cache? **A:** So a fresh value written after our read isn't deleted.

Q: Dining philosophers — who behaves differently under ordering? **A:** The philosopher between forks 4 and 0 takes fork 0 first.

Gotchas

- `if` instead of `while` around waits → spurious-wakeup bugs.
- `notify()` with mixed waiters on one monitor → the wrong thread wakes, hang.
- Tiny Fork/Join tasks → scheduling costs more than work; use a threshold.

TL;DR

Every classic problem = one primitive + one invariant: semaphores as batons, monitors with `notifyAll`, conditions per wait reason, ordered locks, CAS on immutable state, stripes by hash.

14. Case studies: everyday objects

Parking lot, vending machine, coffee machine, ATM, elevator, traffic signal, shipping locker — physical machines with lifecycles, deadlines and contended resources.

Must-know

- **Fit rules live on enums**, spot/slot counts on the owning object, money in **integer cents** or `BigDecimal`.
- **State pattern** when the question is "what can happen next?" (vending, ATM session, traffic signal); **Strategy** for the rule most likely to change (pricing, dispatch, timing).
- **Tick-driven** `step()`/`tick()` makes timed machines deterministic and testable without a clock.
- **Notify outside the lock**; **inject** the lot/machine instead of a Singleton (you can't test two of them otherwise).

Case cards

14.1 Parking lot

- **Requirements:** multiple floors, entries and exits; ticket at entry, pay at exit/portal/attendant (cash or card); never exceed capacity ("full" boards); spot types incl. electric; vehicle types; per-floor display board; tiered hourly fees.
- **Core classes:** `ParkingLot`, `ParkingFloor`, `ParkingSpot`, `Vehicle`, `Ticket` (ACTIVE → PAID → EXITED), `DisplayBoard`, `SpotType`/`VehicleType` enums.
- **Key patterns:** Strategy (pricing, spot-finding), Factory (electric spots), Observer (boards); Singleton rejected — inject the lot.
- **Tricky part:** two gates must never get the same spot → free spots per type in a **concurrent sorted set**, atomic `pollFirst` + CAS on the spot (O(log n), no scan); per-ticket lock for payment; `putIfAbsent` for plates.

14.2 Vending machine

- **Requirements:** slots with code, one product, count; price in cents; choose then pay; reject unknown coins; exact change or refuse and refund everything; cancel with full refund until the item drops; empty slot can't be chosen; operator refills.
- **Core classes:** `VendingMachine`, `Slot`, `Inventory`, `CashBox`, `Denomination`, `VendingState` + Idle, Selected, HasMoney, Dispensing, OutOfStock.
- **Key patterns:** **State is the whole design** — one interface listing every button, no `if` on status.
- **Tricky part:** change-making — money goes into the box on insert, refund = "make exact change for the balance"; greedy fails for non-canonical or short-stocked coin sets → **bounded coin-change DP**; `synchronized` machine.

14.3 Coffee vending machine

- **Requirements:** drinks with price and recipe; menu with availability; pay, change, full refund if the drink can't be made; deduct exactly the recipe; low-stock alerts; refills; customisation.
- **Core classes:** `CoffeeMachine`, `Recipe`, `RecipeBook`, `IngredientStore`, sealed `OrderResult` (Served / Rejected).
- **Key patterns:** recipes as **data** (new drink = new row); Observer for low stock (once per threshold crossing).
- **Tricky part:** two orders take the last milk → **one lock around check-and-deduct of every ingredient** (all-or-nothing, never negative); an `AtomicInteger` per ingredient allows partial deductions; notify outside the lock.

14.4 ATM

- **Requirements:** card + PIN verified **by the bank**; balance, cash deposit, check deposit, withdraw, transfer; card kept all session; journal; checks held until verified; operator refills; manager reports.
- **Core classes:** `Atm`, `AtmState` + 4 states, sealed `Transaction`, `NoteHandler`, `CashDispenser`, `BankService`, `Account` (total vs available balance).
- **Key patterns:** State (session), Command (transactions with one `execute`), Chain of Responsibility (note cassettes).
- **Tricky part:** withdrawal = **plan notes → atomic debit at the bank → dispense → reverse on hardware failure**, journaled; per-account lock at the bank; transfers lock accounts **in id order**; PIN attempts limited at the bank (HSM, encrypted PIN block).

14.5 Elevator system

- **Requirements:** many floors and cars; hall and car panels; a hall call picks one car; serve stops ahead before turning; never move with doors open; max load; out-of-service cars; fire recall.
- **Core classes:** `Elevator`, `StopList` (two sorted sets: up, down), `Door`, `Dispatcher`, `DispatchStrategy`, `Building`.
- **Key patterns:** Strategy (dispatch: nearest suitable car) + modes; LOOK scheduling.
- **Tricky part:** **LOOK** — keep moving while stops remain ahead, reverse at the last request (SSTF can starve a far floor); `step()` does exactly one action (close doors / refuse overload / move one floor / stop and open); safety overrides scheduling; per-car lock + synchronized dispatcher; pending queue when no car is free.

14.6 Traffic signal control

- **Requirements:** multiple roads; red/yellow/green; configurable, traffic-adaptive durations; green → yellow → all-red → next green; emergency priority; fault → all flashing red.
- **Core classes:** `Intersection`, `TrafficLight`, `Phase` (approaches green together), `SignalState` + Green, Yellow, AllRed, EmergencyHold, FlashingRed, `TimingPolicy`, `SafetyMonitor`.
- **Key patterns:** timed State, Strategy (timing), Observer (lights + conflict monitor).
- **Tricky part:** safety invariant — never conflicting greens, not even for an instant; an emergency only **shortens green**, yellow and all-red always run in full; synchronized `tick()` driven by a 1-second timer plus sensor/emergency threads. Not a Singleton: one per intersection.

14.7 Shipping locker

- **Requirements:** stations of numbered doors (S/M/L); carrier reserves a door ahead (2-h hold); deposit → one-time code to the customer only; pickup opens one door once; parcel expires after 3 days → courier collects for return; keypad lockout after 5 wrong codes; nearest station with space; doors out of service.
- **Core classes:** `LockerStation`, `Compartment` (FREE/RESERVED/OCCUPIED/OUT_OF_SERVICE), `Booking` (RESERVED → DEPOSITED → PICKED_UP / EXPIRED → RETURNED / CANCELLED), `CompartmentSelector`, sealed `PickupResult`, `LockerListener`, `LockerNetwork`.
- **Key patterns:** Strategy (smallest-fit door), Observer (SMS, outside the lock), two enum state machines, sealed results; clock and randomness injected.
- **Tricky part:** choose-and-mark is atomic under the station lock (50 carriers, 20 doors → 20 distinct bookings); deadlines enforced **lazily** on every call so doors return even if the scheduler is late; codes random, single-use, stored as a salted hash; door (reusable) kept separate from booking (one parcel's history).

In the interview

Q: Why not a Singleton parking lot? **A:** Hidden global, can't test two lots; create one and inject it (if pressed, a holder-class Singleton is thread-safe).

Q: Vending: why not an enum with a switch? **A:** Every new state edits every switch; a State class is forced by the compiler to answer every button.

Q: Coffee: why notify outside the lock? **A:** A slow or re-entrant listener would block all orders or deadlock.

Q: ATM: two ATMs, one joint account? **A:** Check and debit together under the account's lock (or one DB transaction).

Q: Elevator: where's the lock? **A:** Per car and around dispatch; no global lock across cars.

Q: Traffic: why not jump straight to emergency green? **A:** Cars in the junction need yellow and all-red to clear it.

Q: Locker: what frees a door if the scheduler is down? **A:** Lazy expiry — every call first applies overdue holds and pickup windows; the sweep only sends messages.

Gotchas

- One subclass per spot type → enum + field.
- Greedy change → can fail; DP.
- `double` money in machines → integer cents.
- Global lock over the whole lot → per floor/type or CAS.

TL;DR

Enums for fit rules, State for lifecycles, Strategy for changing rules, atomic claims (pollFirst + CAS, one lock per recipe/account), deterministic ticks, listeners after the lock, no Singletons.

15. Case studies: booking (1) — library, movie tickets, concerts, hotel

Every booking problem is "name the real unit of inventory, then make *if free then take* atomic".

Must-know

- **Unit of inventory** — copy (`BookItem`), show seat (`ShowSeat`), seat/GA count, room-night.
- **Hold → pay → commit** — all-or-nothing hold with **TTL**, payment via a gateway interface, commit re-validates atomically, **refund (compensation)** if the hold was lost.
- **Across servers** — conditional `UPDATE`/unique constraints are the real guarantee; Redis TTL keys for holds.
- **Optimistic vs pessimistic** — pessimistic/conditional update for hot inventory; optimistic when conflicts are rare.

Case cards

15.1 Library management

- **Requirements:** search by title/author/subject/date; many copies per title with barcode + rack; who has this copy / what does this member have; **max 5 books, 10 days**; fines; reservations; notifications (available, overdue).
- **Core classes:** `Book` (title, ISBN) vs `BookItem` (copy), `Member`, `BookLending`, `LendingService`, `Catalog` (one index per search).
- **Key patterns:** Strategy (`FinePolicy`), notifier interface; FIFO reservation queue per title.
- **Tricky part:** the last copy lent twice / a member going over 5 → checkout locks **member then copy** (fixed order); returned copy handed to the title's waitlist head.

15.2 Movie tickets (BookMyShow)

- **Requirements:** cities, cinemas, halls, shows; search; seating plan, pick several seats; no two customers reserve the same seat; card or cash; coupons; notifications; hold expiry.
- **Core classes:** City → Cinema → Hall → `HallSeat`; `Show` owns a `ShowSeat` per hall seat; `Booking`, `SeatLockProvider`, `BookingService`, `CancellationPolicy`.
- **Key patterns:** Strategy (refund rule), gateway behind an interface.
- **Tricky part:** book `ShowSeat`s, not hall seats; `SeatLockProvider` holds all requested seats atomically with a ~5-minute TTL; commit re-checks the hold and refunds if lost; per-show monitor in one JVM, conditional update/unique constraint across servers.

15.3 Concert tickets

- **Requirements:** layout and availability; choose seats or a GA quantity; never oversell; fair access; per-user caps; self-expiring holds; waitlist for sold-out events.
- **Core classes:** `SeatedInventory`, `GaPool`, sealed `SeatState`, `Hold`, `VirtualWaitingRoom`, `PurchaseLimiter`.
- **Key patterns:** sealed seat states, FIFO waitlist, lock-free claims.
- **Tricky part:** extreme contention → `AtomicReference<SeatState>` per seat claimed by **CAS**, GA as an `AtomicInteger` decremented in a CAS loop (never negative), FIFO virtual waiting room, atomic per-user limit, 10-minute holds + a reaper, released tickets go to the waitlist first; queue service down → fail closed.

15.4 Hotel management

- **Requirements:** book room types; search available rooms by dates; who booked what; cancel with full refund > 24 h before check-in; notifications; housekeeping log; room service, amenities; card/check/cash.
- **Core classes:** `Hotel`, `HotelLocation`, `Room`, `RoomBooking`, `Invoice`, `RoomCharge` subclasses, `Payment` subclasses, `FrontDesk`.
- **Key patterns:** Template Method (`RoomCharge` adds an invoice line); payment subclasses.
- **Tricky part: "a status field is not availability"** — each room keeps booked nights; `synchronized Room.tryReserve(nights)` checks and books atomically; the refund rule lives in one method; check-out totals the invoice and kills the keys.

Key facts & numbers

Problem	Contended unit	Atomic step
Library	Copy + member quota	Lock member, then copy
BookMyShow	`ShowSeat`	All-or-nothing TTL hold; atomic commit
Concert	Seat / GA counter	CAS per seat; CAS loop on counter
Hotel	Room × night	`synchronized Room.tryReserve`

In the interview

Q: BookMyShow: why TTL on holds? **A:** Otherwise abandoned checkouts block seats forever.

Q: Concert: lock-free or locks? **A:** Lock-free wins under high contention with tiny critical sections; locks are simpler when many things change together.

Q: One person with 50 accounts? **A:** Verified identity, card fingerprinting, limits on those.

Q: Hotel: two rooms in one booking? **A:** A known limitation of the base design; extend with a `Reservation` holding several `RoomStay`s, all-or-nothing by locking rooms in room-number order (or one DB transaction).

Gotchas

- Booking the physical seat instead of the show seat → double booking across shows.
- Checking availability and booking in separate calls → race.
- Charging before the hold is confirmed, or never refunding a lost hold → compensation path.
- Status field as availability → dates/nights decide.

TL;DR

Name the real unit (copy, show seat, room-night); hold all-or-nothing with TTL → pay → commit atomically → refund if lost; per-item locks in fixed order, CAS under extreme contention, DB conditional updates across servers.

16. Case studies: booking (2) — car rental, restaurant, airline, course registration

Same booking skeleton; availability is about **time** and rules live on the object that owns the data.

Must-know

- **Availability = dates/slots, status = where it is now** (car, table, room).
- **Two-level distinctions** — route vs departure (`Flight`/`FlightInstance`), chair vs chair-on-departure (`Seat`/`FlightSeat`), catalogue vs offering (`Course`/`Section`).
- **Every charge is a bill line**; the check/bill is **computed**.
- **Rules as small objects** (chain) when rules come and go.

Case cards

16.1 Car rental

- **Requirements:** rent cars/trucks/SUVs/vans/motorcycles by barcode; who has this vehicle / what has this member rented; late fees; search and reserve; notifications (pick-up, due, overdue); cancel; vehicle log; insurance,

equipment, services add-ons.

- **Core classes:** `Vehicle`, `DateRange`, `Reservation`, `Bill` / `BillItem`, `ReservationAddOn`, `PricingStrategy`.
- **Key patterns:** Template Method (add-ons), Strategy + Decorator (pricing), reservation state machine.
- **Tricky part:** availability is about **dates** → each vehicle keeps booked days and `synchronized Vehicle.tryBook(range)` checks and books atomically (you can book next month while it's out today); late fee = days late × rate.

16.2 Restaurant management

- **Requirements:** branches, menu per branch (sections → items); waiter creates an order per table with a meal per seat; walk-ins and reservations; cancel; reminders; card/cheque/cash; kitchen queue, waiter notified when ready.
- **Core classes:** `Menu` / `MenuSection` / `MenuItem`, `Table`, `Reservation`, `Order`, `Meal` / `MealItem`, `Kitchen`, `Check`.
- **Key patterns:** Observer ("order ready"), Strategy (chef assignment, payment), order state machine.
- **Tricky part:** two receptionists, one table → table has **status (now) + held slots (later)**; best-fit search, `synchronized Table.tryHold(time)`; listeners after the lock; price lives only on `MenuItem`.

16.3 Airline management

- **Requirements:** search by date/source/destination; itineraries with several passengers; seat maps; admin adds aircraft, flights, schedules, cancels flights (everyone notified); crew assignment; payment; notifications.
- **Core classes:** `Flight` vs `FlightInstance`, `Seat` vs `FlightSeat`, `FlightReservation`, `Itinerary`, `FareStrategy`.
- **Key patterns:** Strategy + Decorator (fares, last-minute surcharge), Facade.
- **Tricky part:** multi-passenger booking gets all seats or none → `FlightInstance.holdSeats` under the instance lock; an **instance → reservations index** so a cancelled flight reaches every passenger (not just the crew).

16.4 Course registration

- **Requirements:** register for sections; capacity with waitlist; prerequisites, credit limit, time conflicts, one section per course, registration window; on drop, promote the waitlist head and notify.
- **Core classes:** `Course` vs `Section` (capacity, slot, bounded FIFO waitlist), `Student`, `RegistrationRule`s, `Registrar`.
- **Key patterns:** chain of rule objects / Specification.
- **Tricky part:** overselling a section → seats taken only inside `Section.enroll`; lock order always **student then section**; drop → promote head → re-check rules → notify; at scale a DB conditional update (09:00 "thundering herd").

In the interview

Q: Car rental: can a car rented today be booked for next month? **A:** Yes — availability is per date range, not a status.

Q: Airline: how do passengers learn of a cancellation? **A:** Through the instance → reservations index.

Q: Course: add a "majors only" rule? **A:** One new `RegistrationRule` class added to the chain.

Q: Restaurant: why a meal per seat (`Map<seat, Meal>`)? **A:** A plate per guest, and split bills become trivial — a flat list of dishes loses that.

Gotchas

- Status as availability → overlapping bookings slip through.
- Booking a `Flight` instead of a `FlightInstance` / `FlightSeat` → one seat sold on every date.
- Rules hard-coded in the registrar → rule objects.
- Prices copied into many places → price on the item; the bill copies it at order time.

TL;DR

Dates, not status; distinguish template from occurrence (flight/instance, course/section); all-or-nothing holds under the owning object's lock; charges are bill lines; rules as small objects.

17. Case studies: games — blackjack, chess, tic-tac-toe, snake & ladder, 2048

Games test rule modelling, polymorphism and deterministic testing (inject randomness).

Must-know

- **Keep generic parts generic** — `Card` / `Deck<T>` know nothing about blackjack values.
- **Inject randomness** (seeded `Random`, stacked shoe, loaded dice, scripted players) so demos and tests repeat.
- **Players and rules as Strategies**; one game = one thread of turns; many games in a `ConcurrentHashMap` with a per-game lock.

Case cards

17.1 Blackjack + deck of cards

- **Requirements:** ante; deal two each (dealer up + hole card); insurance when dealer shows an ace (half the ante, pays 2:1) and dealer peek; hit / stand / double down / split (no re-split); bust = loss; dealer follows a fixed rule; blackjack pays 3:2, win 1:1, push.
- **Core classes:** `Card`, `Deck<T>`, `Shoe`, `Hand`, `BlackjackScoring`, `PlayerStrategy`, `DealerPolicy`, `BlackjackGame`.
- **Key patterns:** Strategy (player decisions, dealer rule); injected `Random` + stacked shoe for tests.
- **Tricky part:** ace = 1 or 11 → hard total (aces as 1) + 10 if an ace fits = **soft** total; blackjack = original two cards totalling 21; value lives in scoring, not on `Card`; whole-chip `int` money works because an even ante keeps payouts whole.

17.2 Chess

- **Requirements:** two players, white first; own pieces only; reject moves that leave your king in check; castling, en passant, promotion; check, checkmate, stalemate; resignation and forfeit; move log.
- **Core classes:** sealed `Piece` + subclasses, `SlidingPiece` (rook/bishop/queen loop), `Board` (8×8), `Move`, `Rules`, `Game`.
- **Key patterns:** polymorphic `targets()`, Factory Method, Command + Memento (undo).
- **Tricky part:** `Rules` = pseudo-legal moves → add specials (using history in `Game`) → **drop moves leaving own king attacked**; no legal move: in check = checkmate, else stalemate; `synchronized` game so a move and a resignation don't interleave.

17.3 Tic-tac-toe

- **Requirements:** N×N board, 2+ players with unique marks; validate inside and empty; detect win or draw after each move.
- **Core classes:** `Board` (row/column/diagonal counters), `Game`, `Player`, `MoveStrategy`.
- **Key patterns:** Strategy (human, seeded random, rule-based "win if possible, else block").
- **Tricky part:** **O(1) win check** — per-player counters for each row, column and both diagonals; a move bumps at most four; reaching N wins (N−1 = "would win"). K-in-a-row needs a directional walk instead.

17.4 Snake and ladder

- **Requirements:** board (~100) with snakes and ladders; several players; roll and move; overshoot rule; optional bonus six with three-sixes penalty; many independent sessions.
- **Core classes:** `Board` (size + `Map<start, Jump>`), `Jump`, `Dice`, `Rules`, `OverflowRule`, `WinStrategy`, `Game`.
- **Key patterns:** Strategy (dice, overflow — exact or bounce, win — first or rank all).
- **Tricky part:** validate the board **at construction** (no jump on the last square, no chains/loops); queue-based turns with a `maxTurns` guard; each game its own `Random`; events to a listener, not `System.out`.

17.5 2048

- **Requirements:** N×N with two starting tiles; four directions slide and merge once; spawn a 2 or 4 only if the board changed; WON (can continue) and LOST detection.
- **Core classes:** `LineMerger`, `Board`, `TileSpawner`, `Game`, `Direction`.
- **Key patterns:** Strategy (spawner, e.g. an "evil" one for tests); one pure function.
- **Tricky part:** write merge **once** — a pure function slides one line toward index 0 (skip zeros, merge equal neighbours once, score); map each direction to lines; spawn 2 (90%) or 4 via a seeded spawner; LOST = no empty cell and no equal neighbours.

Key facts & numbers

Game fact	Value
Blackjack payout / insurance	3:2 / 2:1 (insurance = half the ante)
Tic-tac-toe counters touched per move	≤ 4 (row, column, 2 diagonals)
2048 spawn	2 with 90%, else 4

In the interview

Q: Why not `int value` on `Card`? **A:** Value is game-specific; keep it in the game's scoring.

Q: Chess: where does castling live? **A:** In `Rules`; a `MoveType.CASTLE_*` tells `apply` to move the rook too.

Q: Chess undo with promotion? **A:** The memento holds the old board with the pawn.

Q: How do you test random games? **A:** Seeded `Random`, stacked shoe, loaded dice, scripted strategies.

Gotchas

- Win check scanning the whole board → counters.
- Four copies of 2048 merge logic → one `LineMerger`.
- Unseeded `Random` → non-repeatable demos.
- Validating snake/ladder boards at play time → at construction.

TL;DR

Generic cards, scoring in the game; polymorphic pieces + legality filter; O(1) counters; board validated once; one pure merge function; strategies for players and randomness.

18. Case studies: platforms (1) — Stack Overflow, Cricinfo, LinkedIn, Facebook

Platform problems = rich domain model + roles + feed + notifications + search + reputation.

Must-know

- **Store *who*, not *how many*** — likes/votes as a set or voter map; counts are derived (a counter can't block a repeat or undo one).
- **Derive, don't store** — stats and counts computed from the underlying records so they can't drift.
- **Two-sided operations live in a service** (accepting a connection changes two members); one-sided ones (follow) are just a set.
- **Notifications** — the domain raises an event; a notification service with pluggable channels delivers it (Observer + Strategy).
- **Search** — an inverted index (word/tag → list), never a scan.
- **Privacy / visibility in one place**, called by every read path.

Case cards

18.1 Stack Overflow

- **Requirements:** guests search/view; members ask, answer open questions, comment, vote, flag; bounties; badges; close/reopen and delete votes; tags; one vote per member per post; no voting on own posts.
- **Core classes:** sealed `Post` → `Question` / `Answer` / `Comment`, `Member` hierarchy (incl. `Moderator`), `Tag` (counts its uses), `Bounty`, `QuestionIndex`, facade.
- **Key patterns:** Strategy (reputation rules), Observer (badges on reputation change), Facade; inverted index; privilege table by reputation.
- **Tricky part:** votes stored **per voter** → blocks double voting and lets a changed vote reverse reputation; reputation awarded by the facade asking the policy (posts know nothing about scoring); `synchronized` post for concurrent votes, `AtomicInteger` reputation.

18.2 Cricinfo

- **Requirements:** teams and matches; live ball-by-ball commentary; international rules; squad per tournament, playing eleven per match; player/match/tournament stats; global stat questions; ODI, Test, T20.

- **Core classes:** `Match` (abstract; formats override overs per innings and innings per match) → `Innings` → `Over` → `Ball`; `Team` ⊇ squad ⊇ `Playing11`; `StatQuery`.
- **Key patterns:** Template Method (stat queries), Observer (live pages).
- **Tricky part:** model the **rules**: an over = six **legal** balls (wide/no-ball re-bowled); the innings decides over end, strike and innings end; run outs aren't the bowler's, byes aren't the batter's; stats derived from balls so corrections propagate; one scorer lock per match, observers in `CopyOnWriteArrayList`.

18.3 LinkedIn

- **Requirements:** profiles (experience, education, skills); search members/companies/jobs; send/accept/reject connections; recommendations; posts, comments, likes/shares; messages; notifications; company pages and jobs.
- **Core classes:** `Member`, `Profile` (composed of small parts), `SocialContent` (posts and comments share likes), `ConnectionInvitation`, `ConnectionService`, `SearchIndex`, `FeedService`.
- **Key patterns:** State (invitation), Strategy (feed ranking, notification channels), Facade.
- **Tricky part:** a connection must appear on **both** members → accept lives in `ConnectionService` and updates both under one lock; degrees via **BFS to depth 3**; 2nd-degree badge = your cached 1st-degree set intersects theirs; `connectionCount` is `connections.size()`.

18.4 Facebook

- **Requirements:** profiles; search members/groups/pages/posts; friend requests; follow without friendship; groups and pages; posts with privacy lists; comments, likes, shares; messages; page recommendations; notifications.
- **Core classes:** `Member`, sealed `Audience` (public, friends, custom list, only me), `PrivacyList`, `FriendService`, `VisibilityService`, `NewsFeedService`.
- **Key patterns:** Strategy (feed ranker), Observer (notifications).
- **Tricky part:** **privacy never leaks** — one `VisibilityService.canSee` (post's audience AND the friend graph) called by feed, search, comments, likes and shares; `FriendService` owns the symmetric graph; suggestions = friends-of-friends by mutual count with a second sort key so lists don't reshuffle.

In the interview

Q: Why a set of likes instead of a counter? **A:** A number doesn't record *who*, so it can't block a repeat or undo one; the set gives the count for free.

Q: Why not store `wickets` on `Player`? **A:** It drifts from the ball record and can't be filtered by format; derive it.

Q: Scale Stack Overflow search? **A:** Keep the `Search` API, replace `QuestionIndex` with a search engine fed by events.

Q: How could a private post leak? **A:** Any read path that forgets privacy — search, feed, comments, shares, notifications.

Gotchas

- Storing only `voteCount` → no duplicate protection, no reversal.
- Reputation logic inside `upvote` → policy via the facade.
- Accept updating one side → one-way friendship; service updates both.
- Privacy checked only in the feed → leaks through search/comments.

TL;DR

Share behaviour in a content base class, store who not how many, derive numbers, two-sided operations in services, Observer + Strategy for notifications, inverted indexes for search, one visibility check for every read path.

19. Case studies: platforms (2) — Amazon, stock brokerage/exchange, auction, Spotify, food delivery, grocery POS

Commerce and marketplaces: reservations of stock, money and riders, order state machines, matching.

Must-know

- **Checkout** = reserve all lines atomically → copy into an immutable order → pay (Strategy) → publish (Observer).

- **Order state machine** — illegal transitions have no edge (no cancel after ship).
- **Money reserved at placement** so ten $1,000 orders can't all pass a $1,000 check.
- **One lock per listing / symbol / zone**; notify after unlocking; inject the clock.

Case cards

19.1 Amazon (online shopping)

- **Requirements:** members sell products; anyone searches by name/category and views; only members buy; cart add/remove/change; checkout whole cart; reviews; cancel unshipped orders; notifications on status changes; card or bank transfer; shipment tracking.
- **Core classes:** `Product`, `ShoppingCart`, `Order`, `OrderLog`, `Payment`, `Shipment`, `Customer` → `Guest` / `Member`.
- **Key patterns:** Strategy (payment), Observer (notifications), order state machine, Facade.
- **Tricky part:** last unit, two buyers → `synchronized Product.reserve`, all-or-nothing multi-line reservation; only `Member` can order (LSP — guests have no order method); `BigDecimal` money; double-click checkout → client idempotency key returns the existing order.

19.2 Stock brokerage + matching engine

- **Requirements:** buy/sell; multiple watchlists; market, limit, stop-loss, stop-limit orders; multiple lots per stock; quarterly and yearly tax statements; funding by check/wire/transfer; execution notifications. Engine (Navi): match each new order against the opposite side, lower sells/higher buys first, earlier first at the same price; trade when buy ≥ sell at the **sell price**.
- **Core classes:** abstract `Order` → Market/Limit/StopLoss/StopLimit, `OrderPart` (fills), `StockLot`, `OrderBook`, `MatchingEngine`.
- **Key patterns:** Observer (fills via `OrderListener`), Template Method (statements), pluggable trade-price policy.
- **Tricky part:** the four order types differ in exactly two things — **may I trade now?** and **worst acceptable price**; book per symbol = two `TreeMap<price, FIFO queue>`: best level O(log L), fill O(1); one lock per symbol; lots sold FIFO.

19.3 Online auction

- **Requirements:** create listings; search by name/category/price range; bid on active listings; keep and announce the highest bid; close at end time — highest bidder wins if the reserve is met; many simultaneous bids stay consistent.
- **Core classes:** `AuctionListing`, `Bid`, `BidResult`, `AuctionService`, `AuctionListener`.
- **Key patterns:** Observer, Specification (search), status state machine `SCHEDULED → ACTIVE → CLOSED | CANCELLED`.
- **Tricky part:** all bid rules (seller, current leader, time, minimum increment) in **one per-listing critical section** — simpler than CAS because several checks are involved; refusals are values with the next minimum, not exceptions; closing is idempotent; notify outbid users after unlocking.

19.4 Music streaming (Spotify)

- **Requirements:** browse/search songs, albums, artists; playlists; play, pause, skip, seek, previous, shuffle, repeat; recommendations from history; FREE vs PREMIUM limits.
- **Core classes:** `Song` / `Album` / `Artist` (immutable records), `MusicLibrary`, `Playlist`, `MusicPlayer`, `PlaybackOrder`, `Recommender`.
- **Key patterns:** Strategy (play order, recommendations), Observer (plays → history), plan rules in an enum.
- **Tricky part:** player = queue + order + cursor + repeat mode; shuffle is a **permutation** rebuilt around the current song (no repeats; previous walks back along the same order); two devices pressing "next" → `synchronized` player methods.

19.5 Food delivery (Swiggy) + FoodKart

- **Requirements:** browse restaurants and menus, order, track with notifications, several payment methods; restaurants manage menus; agents accept and deliver. FoodKart: register/login users, owner registers a restaurant (one dish, many pincodes), update quantity/location, rate 1–5, show restaurants sorted by rating or price, place orders.
- **Core classes:** `Restaurant`, `Cart` (one restaurant), `Order`, `DeliveryAgent`, `DeliveryAssignmentStrategy`, `RestaurantSelectionStrategy`.
- **Key patterns:** Strategy (assignment, selection, payment), Observer, order state machine PENDING → … → DELIVERED.

- **Tricky part:** kitchen capacity reserved atomically on accept and released on "ready"; riders claimed with **CAS** so none gets two orders; the strategy returns a ranked list so you fall through if the top rider is claimed first; FIFO queue when no rider is free; scale by zone with one dispatcher per zone.

19.6 Grocery store (POS & checkout)

- **Requirements:** scan barcodes, weigh produce (PLU + grams), void lines, item offers (buy X get Y, multi-buy, % off department) + basket coupon, tax per category (inclusive or exclusive), split tender (cash with change, card, UPI), receipt, returns against the receipt, stock with low/negative alerts, several registers.
- **Core classes:** `Register` (facade), `Sale` (OPEN → TENDERING → COMPLETED / CANCELLED), `LineItem` (price copied at scan), `PricingEngine` → `PriceBreakdown`, `ItemPromotion`/`BasketPromotion`, `TaxPolicy`, sealed `Tender`, `Receipt`, `ReceiptStore`, `Inventory`.
- **Key patterns:** Strategy (offers, tax, payment gateway), Facade, Observer (stock), enum state machine, sealed tenders.
- **Tricky part:** a **pure** pricing engine: best single item offer per SKU → one basket coupon → spread it over SKUs (remainder to the largest) so per-category tax and per-item refunds are exact; prices frozen at `total()`; refunds = net paid, claimed atomically with `compute`; stock via atomic `merge`, allowed to go **negative** (the item is in hand), edge-triggered alerts.

In the interview

Q: Amazon: tax and discounts? **A:** A `PricingRule` chain (CoR or list of Strategies) applied when building the order; the order stores final line prices.

Q: Matching engine trade price? **A:** Navi problem: the sell price. Real continuous trading: the resting order's price.

Q: Auction: why not `AtomicReference` + CAS? **A:** The rule also checks seller, leader and time; one small lock per listing is simpler and still fast.

Q: Auction close on time across servers? **A:** Scheduler with a time-ordered index; `closeIfDue` is idempotent.

Q: Food delivery: why release kitchen capacity on "ready"? **A:** The kitchen is free once food leaves; delivery is the rider's constraint.

Q: Grocery: two offers match the same product? **A:** Apply only the one worth most to the customer; then the basket coupon on what is still owed — a written stacking rule, not list order.

Gotchas

- Guests with an `order()` that throws → LSP; ordering only on `Member`.
- Checking funds at placement without reserving → over-commitment.
- Notifying bidders inside the lock → slow listeners block bidding.
- "Random next" as shuffle → repeats; use a permutation.

TL;DR

Reserve atomically, copy into immutable orders, pay via Strategy, publish via Observer; state machines for orders and listings; per-symbol order books with price-then-time priority; CAS for riders; lock per listing; notify after unlocking.

20. Case studies: components — Splitwise, notifications, cache/LRU, Uber, logger, pub-sub, rate limiter

Reusable components where one or two patterns carry the whole design.

Must-know

- **One mutator / one exit path** — ledger `adjust`, cache `remove`, notification `finish`: invariants and stats live in one place.
- **Strategy for the rule most likely to change** (split, send rule, eviction, matching, pricing, formatter, limiter algorithm).
- **Decorator wraps a rule and stays in the call path** (surge pricing, async appender); Strategy *replaces* it.
- **At-least-once + idempotency** whenever retries exist.

Case cards

20.1 Splitwise

- **Requirements:** users, groups, expenses split equal / exact / percentage; shares must add up exactly or nothing changes; balance with one person and overall; settle up; simplify debts; 2-decimal money, no lost cents.
- **Core classes:** `Expense`, `SplitStrategy` (Equal/Exact/Percent), `Share`, `Ledger`, `DebtSimplifier`.
- **Key patterns:** Strategy (splits).
- **Tricky part:** `Ledger` keeps **one signed number per pair** with a single `adjust`, so debts cancel automatically and settle-up is a reverse adjust; simplification = net positions + greedy biggest-debtor-pays-biggest-creditor, ≤ n − 1 payments (optimal is NP-hard); `BigDecimal` with explicit rounding.

20.2 Notification service

- **Requirements:** email, SMS, push; per-user channel preference **per kind**; templates; quiet hours (may wrap midnight; OTP ignores them); rate limit per user per kind per hour; bounded retries then fall back to the next channel, all fail → FAILED; audit every attempt; caller doesn't wait.
- **Core classes:** abstract `NotificationChannel` + channels, `SendRule`, `ChannelFactory`, `DeliveryListener`.
- **Key patterns:** Template Method (channel `deliver` holds retry-with-backoff), Strategy (send rules returning a reason), Observer (outcomes), Simple Factory, Adapter (providers).
- **Tricky part:** render once, try channels in order, fall back, one `finish()` exit; rate-limit check-then-act gap; duplicates on retry → idempotency keys, at-least-once.

20.3 In-memory cache / LRU

- **Requirements:** get/put in O(1); `put` overwrites; evict before inserting when full; LRU by default, swappable FIFO/LFU; optional TTL (expired read = miss); invalidate/clear; stats and removal listeners; thread-safe.
- **Core classes:** `Cache<K,V>`, `Node`, `DoublyLinkedList`, `EvictionPolicy`, `Cache.Builder`, `ShardedCache`.
- **Key patterns:** Strategy (eviction), Observer (removals), Builder (config).
- **Tricky part:** `HashMap` + doubly linked list with sentinels; the node stores its **key** so eviction can delete it from the map; in LRU a `get` is a write → one lock (or shard); listeners outside the lock; lazy TTL + sweeper; O(1) LFU via frequency buckets.

20.4 Ride sharing (Uber)

- **Requirements:** economy/premium/van; offer to online, free drivers of the right type in range, nearest first, decline → next; trip requested → accepted → started → ended, cancel only before start; fare = base + per km + per minute with a minimum, surge multiplies it; ratings; payment.
- **Core classes:** `Rider`, `Driver`, `Trip`, `MatchingStrategy`, `PricingStrategy`, `SurgePricing`.
- **Key patterns:** Strategy ×3 (matching, pricing, payment), Decorator (surge), trip state machine.
- **Tricky part:** a driver can never be on two trips → **atomic claim** (`AtomicBoolean` CAS); ranked list + cursor for declines; surge **wraps** the pricing rule instead of a flag; ratings = total + count; `BigDecimal` fares. Carpool variant: users offer seats, riders select by Most Vacant / Preferred Vehicle.

20.5 Logging framework

- **Requirements:** levels TRACE < DEBUG < INFO < WARN < ERROR < FATAL (+ OFF); named loggers in a dot hierarchy inheriting the nearest ancestor's level; console/file/memory destinations with their own thresholds; pattern and JSON formats; propagation to ancestors unless additivity is off; optional async.
- **Core classes:** `Logger`, `LogManager`, `LogRecord` (immutable), `Appender`, `AsyncAppender`, `Formatter`.
- **Key patterns:** Chain of Responsibility (logger hierarchy), Singleton (`LogManager`), Strategy (formatters), Template Method (appender base), Decorator (async).
- **Tricky part:** a disabled debug call must cost one comparison (use `Supplier` for expensive messages); never crash the app; never block on a full async queue; queued logs are lost on crash unless flushed on shutdown.

20.6 Pub-sub

- **Requirements:** create topics; publish (key, payload) → offset; subscribe from EARLIEST or LATEST; in-order delivery; commit only after the callback returns; retry then dead-letter; seek/replay; unsubscribe; clean shutdown.
- **Core classes:** `Broker`, `Topic` (append-only log), `Subscription` (committed offset), `Subscriber`, `Message`.
- **Key patterns:** Observer through a broker.

- **Tricky part:** one dispatcher thread per subscription (isolation + order; the log is shared, progress isn't), waiting on a condition (no busy-wait); ack after processing = **at-least-once** → idempotent subscribers; exactly-once is hard; with partitions, order is per key.

20.7 Rate limiter

- **Requirements:** `check(userId, api)` → `Decision(allowed, remaining, retryAfter)`; rules per API and tier with a default; one limiter per (user, api) created on first use; pluggable algorithms; combined rules must all allow.
- **Core classes:** `Limiter` (TokenBucket, FixedWindow, SlidingWindowLog, SlidingWindowCounter), `CompositeLimiter`, `LimiterFactory`.
- **Key patterns:** Strategy, Simple Factory, Composite (probe all, commit all), Facade.
- **Tricky part:** atomic refill-check-take; `computeIfAbsent` for one limiter per key; fixed window allows up to **2N** across a boundary; across servers → Redis + Lua.

Key facts & numbers

Algorithm	Behaviour	Memory
Token bucket	Allows bursts; refill at fixed rate	O(1)
Fixed window	Simple; up to 2N at a window edge	O(1)
Sliding window log	Exact	O(N) timestamps
Sliding window counter	Approximate	O(1)

LRU option	get / put	Thread-safe	Use
`HashMap` + own doubly linked list	O(1) / O(1)	With a lock	Interviews, custom behaviour
`LinkedHashMap(16, 0.75f, true)` + `removeEldestEntry`	O(1) / O(1)	No	Quick single-threaded caches
Caffeine / Guava	amortized O(1)	Yes	Production Java

Code to remember

```
synchronized V get(K key) {                          // LRU: get is a write (moves to front)
    Node n = index.get(key); if (n == null) return null;
    unlink(n); addFront(n); return n.value;
}
synchronized void put(K key, V value) {
    Node n = index.get(key);
    if (n != null) { n.value = value; unlink(n); addFront(n); return; }
    if (index.size() == capacity) { Node lru = tail.prev; unlink(lru); index.remove(lru.key); } // node keeps its key
    n = new Node(key, value); addFront(n); index.put(key, n);
}
```

In the interview

Q: Notification: why Template Method for channels, not Strategy? **A:** Channels share one algorithm (retry with backoff) and vary in one step; injecting a `RetryPolicy` + `Sender` strategy is a valid composition alternative.

Q: Uber: Strategy vs Decorator for surge? **A:** Strategy replaces the rule; Decorator wraps it and stays in the call path (holds an `inner` of the same interface).

Q: Pub-sub: slow consumer? **A:** One dispatcher per subscription; others are unaffected.

Q: Token bucket vs sliding log? **A:** Bucket: bursts, O(1) memory; log: exact, stores every timestamp.

Q: Splitwise: why is minimum-transfer simplification not guaranteed? **A:** Optimal is NP-hard; greedy gives at most n − 1 payments.

Gotchas

- Several code paths that change balances or remove cache entries → one mutator / one exit.
- `double` in Splitwise → lost cents; `BigDecimal` + explicit rounding.

- Surge as a boolean flag → Decorator.
- Blocking the app on a full log queue → drop or fall back, never block.

TL;DR

Ledger = one signed number per pair; notifications = template channels + send-rule strategies + one `finish`; LRU = map + list, get is a write; Uber = strategies + surge decorator + CAS claim; logger = hierarchy chain + async decorator; pub-sub = log + offsets + one dispatcher per subscription; rate limiter = strategies + composite + atomic refill-check-take.

21. Machine-coding round problems

The 90-minute format: working, demo-able, layered code that handles bad input.

Must-know

- **Packages** — model, repository, service, strategy, exception, driver.
- **Driver** — the statement's sample input as a `List<String>` of commands; `Map<String, Consumer<String[]>>` command processor; catch domain exceptions per command, print and continue.
- **Services** return data and throw domain exceptions; **no `System.out` in services**.
- **Repository** = interface + `ConcurrentHashMap` implementation; IDs from an `AtomicInteger`.
- **Entities guard their own invariants** (`Task.changeStatus` rejects `TODO → DONE`).
- **Run after every requirement** (at least every 15 minutes); mandatory in order, bonus last.

Case cards

21.1 Task management

- **Requirements:** users; create/update/delete/assign tasks; status transitions; comments; search with any filter combination + sort; reminders; per-task history; thread-safe.
- **Core classes:** `Task`, `TaskStatus`, `TaskService`, `TaskRepository`, `TaskFilter`.
- **Key patterns:** Specification (filters), Strategy (sorts), Observer (notifications), enum state machine.
- **Tricky part:** `Task` guards its own transition table and history; statuses have no behaviour beyond legal transitions, so an **enum table beats the full State pattern**; per-task `synchronized`.

21.2 Digital wallet

- **Requirements:** create wallet; top up; atomic transfer; statement with running balance; cashback offers paid from a PROMO wallet; idempotent top-up and transfer.
- **Core classes:** `Wallet`, `Transaction`, `Money`, `Offer`, `WalletService`.
- **Key patterns:** Strategy (offers), idempotency key.
- **Tricky part:** lock both wallets in a **global (id) order**, debit before credit, idempotency key → at most once; total money conserved (only top-ups add money) — prove it with a concurrent demo printing `conserved=true`.

21.3 Event calendar (Flipkart)

- **Requirements:** users with working hours; teams; create an event only if everyone is inside hours, free, and each team has N free members; events in a range; free slots; bonus: atomic reschedule, recurring events.
- **Core classes:** `TimeSlot` (half-open), `Event`, `EventStore`, `RepresentativeSelector`, `CalendarService`.
- **Key patterns:** Strategy (representative choice), plan-then-save.
- **Tricky part:** **half-open intervals** `[start, end)` so back-to-back events don't clash; a side-effect-free `plan()` reused by create, recurring and update; atomic check-and-book; free slots by a boundary sweep; ask about the contradictory sample.

21.4 Coding Blox (Flipkart)

- **Requirements:** CreateUser, CreateQuestion, CreateContest, AttendContest, RunContest, LeaderBoard (+ bonus commands); errors printed, processing continues; scoring and solving replaceable.
- **Core classes:** `Contest`, `Level`, `User`, `Question`, `SolveSimulator`, `CommandProcessor`.
- **Key patterns:** command map, Strategy (solver, scoring); scoring on the `Level` enum.
- **Tricky part:** randomness behind a **seeded** `SolveSimulator`; `Contest` guards status so it can't run twice; atomic rating update if two contests finish at once.

21.5 PropertyHunt (ClearTrip)

- **Requirements:** register, list property, search with any combination of filters, shortlist, view shortlisted/listed/sold, mark sold (+ bonus).
- **Core classes:** `Property`, `SearchCriteria` (+ Builder), `PropertyService`.
- **Key patterns:** Builder + predicate list, Strategy (sort).
- **Tricky part: normalise at the boundary** (`Price.parse("90L")`, `AreaUnit`); each optional criterion becomes a predicate, AND them; sort by (location rank, key, id); shortlists hold ids; synchronized mark-sold.

21.6 Ledger Co (Navi)

- **Requirements:** LOAN, PAYMENT (lump sum), BALANCE commands; reproduce the official samples; bad lines don't stop the rest; interest rule replaceable; exact `BigDecimal`.
- **Core classes:** `Loan`, `LumpSum`, `Balance`, `InterestPolicy`, `LedgerService`.
- **Key patterns:** Strategy (interest), derived state, command loop.
- **Tricky part:** EMI = ⌈(P + P·N·R/100) / 12N⌉; store facts (loan + lump sums), **derive** amount paid and EMIs left on demand with ceiling rounding; compute one sample by hand first.

21.7 Inventory management

- **Requirements:** products (reorder point, max level) and warehouses; receive; ship by strategy; transfer; cycle-count adjust; stock per warehouse and total; movement history; reorder alerts; no overselling.
- **Core classes:** `Product`, `Warehouse`, `StockMovement`, `AllocationStrategy`, `ReorderListener`.
- **Key patterns:** Strategy (allocation), Observer (edge-triggered low-stock alerts).
- **Tricky part:** data shape `stock[sku][warehouse] = qty` + append-only movement ledger; ship = plan-then-apply under a **lock per SKU** (different products in parallel); alert once per dip, not on every shipment below the line.

21.8 Unix file search (`find`)

- **Requirements:** in-memory file tree; tests by name/iname glob, type, size, mtime, owner, empty; combine with AND / OR / NOT and parentheses; min/max depth, limit, prune; parse `find` command lines; clear errors.
- **Core classes:** `Node` → `FileNode` / `DirectoryNode`, `FileSystem`, `FileFilter`, `AndFilter` / `OrFilter` / `NotFilter`, `Filters` (leaf tests), `SearchCriteria` + `Builder`, `FileSearcher`, `FindCommandParser`.
- **Key patterns:** Specification (one class per test), Composite ×2 (file tree, filter tree), Builder (criteria), recursive-descent parser (Interpreter-style).
- **Tricky part:** precedence `!` > AND (implicit) > `-o` — `-name a -o -name b -type f` = `a OR (b AND file)`; GNU `-size` rounds **up** (`-size -1M` = empty files only); iterative DFS with an explicit stack streaming to a `Consumer`; prune ≠ filter (prune never enters the directory); children in a `ConcurrentSkipListMap` (sorted, no CME).

Key facts & numbers

Rubric item	What earns it
Working code	Thin end-to-end path in the first hour; never finish non-compiling
Completeness	Mandatory first, in the statement's order
Clean models	Enums for fixed sets, records for values, classes for changing things
Separation	Driver → Service → Repository
Extensibility	New rule = new class
Edge cases	Custom exceptions; the driver keeps going
Readability	Names from the statement's nouns and verbs
Concurrency (if asked)	Say where the lock is; demo with a pool
Tests (bonus)	`assert`-style self-check in `main`

Code to remember

```
Map<String, Consumer<String[]>> commands = Map.of(
    "CREATE_WALLET", a -> System.out.println(service.create(a[1], new BigDecimal(a[2]))),
    "TRANSFER",      a -> System.out.println(service.transfer(a[1], a[2], new BigDecimal(a[3]), a[4])));
for (String line : sampleInput) {
    String[] a = line.trim().split("\\s+");
    try { commands.getOrDefault(a[0], x -> System.out.println("Unknown: " + x[0])).accept(a); }
    catch (DomainException e) { System.out.println("Error: " + e.getMessage()); }   // print and continue
}
```

In the interview

Q: Why an interface for an in-memory repository? **A:** Services depend on an abstraction; JDBC or Redis can replace it later.

Q: Wallet: how do you avoid deadlock on A→B and B→A transfers? **A:** Always lock the lower wallet id first.

Q: Why half-open time slots? **A:** An event ending at 10:00 and one starting at 10:00 must not conflict.

Q: `find`: why And/Or/Not as classes, not just lambdas? **A:** The parser builds them as a visible tree you can print and test; a new test is one class and the searcher never changes.

Q: What do you do in the last 15 minutes? **A:** Edge cases and validation, thread-safety where money or stock is shared (say where the lock is), clean names, run everything once more, and prepare a 2-minute walkthrough.

Gotchas

- 20 minutes on a perfect class diagram → enums/records first; the diagram lives in the names.
- One `Manager` with 15 maps → a service per area + repositories.
- `System.out` in services → the driver prints.
- `double` money → `BigDecimal` or whole units in a `long`.
- Crashing on the first bad command → catch per command.
- Unseeded `Random`, `LocalDate.now()` in logic → seed it / inject a `Clock`.

TL;DR

Working code first, clean layering second, patterns where the statement asks for variation, graceful errors everywhere; model → repository → service → strategy → exception → driver; run often.

22. Revision: cheat sheets & glossary

The handbook's own one-pagers, reduced to what you check in the last hour.

Must-know

- **Revise by recall**, then check. Six patterns matter most: Strategy, Factory, Observer, State, Singleton, Builder. Always know the "when *not* to use it".
- **Every case study teaches the same five things:** name the real unit of inventory; every "if free then take" is one atomic step; put rules on the object that owns the data; Strategy for the rule most likely to change, Observer for "tell someone", State/enum machine for lifecycles; inject time and randomness, exact money, listeners outside locks.

Key facts & numbers

Edge-case category	Ask your code
Input	`null`, empty, negative, zero, duplicate/unknown IDs
Ownership	Wrong user acting on another's object; moving out of turn
Capacity	Full lot, zero seats, max loans, party larger than any table
State	Paying twice, exit without entry, cancel after delivery, move after game over
Time	Exit before entry, expiring holds, part hours, half-open boundaries, late returns
Money	`BigDecimal` scale/rounding, no lost cents, `compareTo` not `equals`
Concurrency	Two threads, one seat/spot/rider; sorted multi-item locks; idempotent retries
Failure	Payment declined after hold; hold lost → refund; listener throws
Randomness	Seeded or scripted

Glossary term	One line
Compensation	Undo a completed step with a new action (refund)
Idempotency	Doing it twice = doing it once; enforced with a key
Hold with TTL	Temporary claim that expires by itself
Composition root	The one place (`main`) where objects are created and wired
Expression problem	Visitor: cheap new operations, expensive new types (plain polymorphism: the reverse)
Qualified association	Lookup by key, i.e. a `Map`
Rule of three	Abstract at the second or third case, not the first
Two hats	Add function or refactor, never both at once

In the interview

Q: What makes every booking problem safe? **A:** Naming the real unit of inventory and making "if free then take" one atomic step, plus holds that expire.

Q: A `switch (vehicleType)` computes fees — principle, smell, pattern? **A:** OCP; Repeated Switches (if repeated); Strategy.

Gotchas

- Re-reading instead of recalling → close the book, say it, then check.
- Memorising all 23 patterns equally → master the six; know the rest by one row.

TL;DR

Framework → classes by behaviour → SOLID as diagnosis → patterns from signals → concurrency recipe → machine-coding discipline.

Last-minute checklist

- [ ] Say the framework: clarify → requirements (8–12 + **out of scope**) → use cases → entities → relationships → class diagram → flows → patterns/SOLID → code → edge cases/concurrency → extensions.
- [ ] Quarter of the time before naming a class; something running by halfway in machine coding.
- [ ] Different data → field/enum; different behaviour → class; one sentence per class, no "and".
- [ ] Don't forget the record class (ticket, loan, booking) and the rule class (rate, strategy).
- [ ] Diamond at the whole (◆ dies with it, ◇ outlives); triangle at the parent; label every line with multiplicity.
- [ ] SOLID pass: two findings is good; state observation → cost → fix → price.

- [ ] One or two patterns from signals; name one you rejected.
- [ ] State vs Strategy: do the alternatives know each other? Decorator vs Proxy vs CoR: add / guard / may stop.
- [ ] Singleton only if genuinely one; otherwise inject. DCL needs `volatile`; `enum` is most robust.
- [ ] Builder: required in constructor, validate in `build()`, immutable product.
- [ ] Observer: iterate a copy, catch per listener, unsubscribe, notify **after** unlocking.
- [ ] Concurrency: name shared state → invariant → atomic check-then-act → smallest tool → fine-grained locks in **sorted order** → holds with TTL → what changes across servers.
- [ ] `volatile` ≠ atomic; `ConcurrentHashMap` compound ops via `computeIfAbsent`/`merge`; `while` around waits.
- [ ] Coffman: mutual exclusion, hold and wait, no preemption, circular wait.
- [ ] Booking: real unit of inventory (show seat, room-night, vehicle-day); hold → pay → commit → refund if lost.
- [ ] LRU: map + doubly linked list, node stores its key, get is a write.
- [ ] Money in `BigDecimal` (from strings, `compareTo`) or integer cents; time via injected `Clock`; seeded `Random`.
- [ ] Machine coding: model/repository/service/strategy/exception/driver; services return data; driver prints and continues.
- [ ] Edge cases: null/empty/duplicate, wrong owner, full, paying twice, boundaries, two threads, payment failure.
- [ ] Finish with extensions and one defended trade-off.